Math 3020 Exercise Solutions

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Chapter 2

2.1 Out of six computer chips, two are defective. If two chips are randomly chosen for testing (without replacement), compute the probability that both of them are defective. List all the outcomes in the sample space.

Solution. This is a combination (order doesn't matter) without replacement. There are $C(6,2)$ ways to pick any 2 from 6 chips, and $C(2,2)$ ways to pick the 2 ones from the 2 defective chips. By counting, the probability of picking the 2 defective ones from all 6 is

$$P(\text{Both are defective}) = \frac{C(2,2)}{C(6,2)} = \frac{1}{\frac{6\cdot 5}{2\cdot 1}} = \frac{1}{15}.$$

Let D denote a defective chip and N a non-defective one. Then there are three outcomes: DD, DN, and NN, because the order doesn't matter.

2.4 Among employees of a certain firm, 70% know Java, 60% know Python, and 50% know both languages. What portion of programmers

(a) does not know Python?

(b) does not know Python and does not know Java?

(c) knows Java but not Python?

(d) knows Python but not Java?

(e) If someone knows Python, what is the probability that he/she knows Java too?

(f) If someone knows Java, what is the probability that he/she knows Python too?

Solution. Denote

$$J = \text{“knows Java”},$$
$$Y = \text{“knows Python”}.$$

Then $J \cap Y$ means knowing both. Moreover, we have

$$P(J) = 70\%,$$
$$P(Y) = 60\%,$$
$$P(J \cap Y) = 50\%.$$

The answers are as follows.

(a) $P(Y^c) = 1 - P(Y) = 40\%$.

(b) $P(Y^c \cap J^c) = P((Y \cup J)^c) = 1 - P(Y \cup J) = 1 - P(Y) - P(J) + P(Y \cap J) = 20\%$.

(c) $P(J \cap Y^c) = P(J) - P(J \cap Y) = 20\%$ because $J = (J \cap Y) \cup (J \cap Y^c)$ is a disjoint union to form J.

(d) $P(J^c \cap Y) = P(Y) - P(J \cap Y) = 10\%$.

(e) $P(J|Y) = \frac{P(J \cap Y)}{P(Y)} = \frac{50\%}{60\%} = \frac{5}{6}$.

(f) $P(Y|J) = \frac{P(J \cap Y)}{P(J)} = \frac{50\%}{70\%} = \frac{5}{7}$.

2.5 A computer program is tested by 3 independent tests. When there is an error, these tests will discover it with probabilities 0.2, 0.3, and 0.5, respectively. Suppose that the program contains an error. What is the probability that it will be found by at least one test?

Solution. Let A_i be the event that the i-th test finds the error. Then we have

$$
\begin{aligned}
P(A_1) &= 0.2, \\
P(A_2) &= 0.3, \\
P(A_3) &= 0.5.
\end{aligned}
$$

The probability that the error will be found by at least one test is

$$
\begin{aligned}
P(A_1 \cup A_2 \cup A_3) &= 1 - P((A_1 \cup A_2 \cup A_3)^c) \\
&= 1 - P(A_1^c \cap A_2^c \cap A_3^c) \\
&= 1 - P(A_1^c)P(A_2^c)P(A_3^c) \\
&= 1 - (1 - 0.2)(1 - 0.3)(1 - 0.5) \\
&= 1 - (0.8)(0.7)(0.5) \\
&= 1 - 0.28 \\
&= 0.72.
\end{aligned}
$$

Note that we used the De Morgan's law in the second line and the independence of the tests in the third line.

2.6 Under good weather conditions, 80% of flights arrive on time. During bad weather, only 30% of flights arrive on time. Tomorrow, the chance of good weather is 60%. What is the probability that your flight will arrive on time?

Solution. Let G be the event that the weather is good, and B be the event that the weather is bad. Let T be the event that the flight arrives on time. Then we have

$$
\begin{aligned}
P(T|G) &= 0.8, \\
P(T|B) &= 0.3, \\
P(G) &= 0.6, \\
P(B) &= 0.4.
\end{aligned}
$$

By the law of total probability,

$$
\begin{aligned}
P(T) &= P(T|G)P(G) + P(T|B)P(B) \\
&= (0.8)(0.6) + (0.3)(0.4) \\
&= 0.48 + 0.12 \\
&= 0.6.
\end{aligned}
$$

So the probability that your flight will arrive on time is 60%.

2.7 A system may become infected by some spyware through the internet or e-mail. Seventy percent of the time the spyware arrives via the internet, thirty percent of the time via e-mail. If it enters via the internet, the system detects it immediately with probability 0.6. If via e-mail, it is detected with probability 0.8. What percentage of times is this spyware detected?

Solution. Let I be the event that the spyware arrives via the internet, and E be the event that it arrives via e-mail. Let D be the event that the spyware is detected. Then we have

$$\begin{aligned} P(D|I) &= 0.6, \\ P(D|E) &= 0.8, \\ P(I) &= 0.7, \\ P(E) &= 0.3. \end{aligned}$$

By the law of total probability,

$$\begin{aligned} P(D) &= P(D|I)P(I) + P(D|E)P(E) \\ &= (0.6)(0.7) + (0.8)(0.3) \\ &= 0.42 + 0.24 \\ &= 0.66. \end{aligned}$$

So the probability that the spyware is detected is 66%.

2.9 Successful implementation of a new system is based on three independent modules. Module 1 works properly with probability 0.96. For modules 2 and 3, these probabilities equal 0.95 and 0.90. Compute the probability that at least one of these three modules fails to work properly.

Solution. Let M_i be the event that module i works properly. Then we have

$$\begin{aligned} P(M_1) &= 0.96, \\ P(M_2) &= 0.95, \\ P(M_3) &= 0.90. \end{aligned}$$

The probability that all three modules work properly is

$$\begin{aligned} P(M_1 \cap M_2 \cap M_3) &= P(M_1)P(M_2)P(M_3) \\ &= (0.96)(0.95)(0.90) \\ &= 0.864. \end{aligned}$$

Therefore, the probability that at least one module fails is

$$\begin{aligned} 1 - P(M_1 \cap M_2 \cap M_3) &= 1 - 0.864 \\ &= 0.136. \end{aligned}$$

So the probability that at least one module fails to work properly is 13.6%.

2.10 Three computer viruses arrived as an e-mail attachment. Virus A damages the system with probability 0.4. Independently of it, virus B damages the system with probability 0.5. Independently of A and B, virus C damages the system with probability 0.2. What is the probability that the system gets damaged?

Solution. Let V_i be the event that virus i damages the system. Then we have

$$\begin{aligned} P(V_1) &= 0.4, \\ P(V_2) &= 0.5, \\ P(V_3) &= 0.2. \end{aligned}$$

The probability that the system gets damaged is

$$\begin{aligned} P(V_1 \cup V_2 \cup V_3) &= 1 - P(V_1^c \cap V_2^c \cap V_3^c) \\ &= 1 - P(V_1^c)P(V_2^c)P(V_3^c) \\ &= 1 - (1 - 0.4)(1 - 0.5)(1 - 0.2) \\ &= 1 - (0.6)(0.5)(0.8) \\ &= 1 - 0.24 \\ &= 0.76. \end{aligned}$$

So the probability that the system gets damaged is 76%.

2.11 A computer program is tested by 5 independent tests. If there is an error, these tests will discover it with probabilities 0.1, 0.2, 0.3, 0.4, and 0.5, respectively. Suppose that the program contains an error. What is the probability that it will be found

(a) by at least one test?

(b) by at least two tests?

(c) by all five tests?

Solution.

(a) Let A_i be the event that the i-th test finds the error. Then we have

$$\begin{aligned} P(A_1) &= 0.1, \\ P(A_2) &= 0.2, \\ P(A_3) &= 0.3, \\ P(A_4) &= 0.4, \\ P(A_5) &= 0.5. \end{aligned}$$

Therefore, the probability that the error will be found by at least one test is

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5) &= 1 - P(A_1^c \cap A_2^c \cap A_3^c \cap A_4^c \cap A_5^c) \\ &= 1 - P(A_1^c)P(A_2^c)P(A_3^c)P(A_4^c)P(A_5^c) \\ &= 1 - (1 - 0.1)(1 - 0.2)(1 - 0.3)(1 - 0.4)(1 - 0.5) \\ &= 1 - (0.9)(0.8)(0.7)(0.6)(0.5) \\ &= 1 - 0.1512 \\ &= 0.8488. \end{aligned}$$

So the probability that the error will be found by at least one test is 84.88%.

(b) Let X_0 be the event that the error is found by 0 tests, namely,

$$X_0 = A_1^c \cap A_2^c \cap A_3^c \cap A_4^c \cap A_5^c.$$

Hence

$$P(X_0) = P(A_1^c)P(A_2^c)P(A_3^c)P(A_4^c)P(A_5^c) = (0.9)(0.8)(0.7)(0.6)(0.5) = 0.1512.$$

Let X_1 be the event that the error is found by exactly 1 test, and Y_i be the event that the error is found by the i-th test and not by any other tests, namely,

$$X_1 = \bigcup_{i=1}^{5} Y_i,$$

where

$$\begin{aligned}
Y_1 &= A_1 \cap A_2^c \cap A_3^c \cap A_4^c \cap A_5^c,\\
Y_2 &= A_1^c \cap A_2 \cap A_3^c \cap A_4^c \cap A_5^c,\\
Y_3 &= A_1^c \cap A_2^c \cap A_3 \cap A_4^c \cap A_5^c,\\
Y_4 &= A_1^c \cap A_2^c \cap A_3^c \cap A_4 \cap A_5^c,\\
Y_5 &= A_1^c \cap A_2^c \cap A_3^c \cap A_4^c \cap A_5.
\end{aligned}$$

We see that

$$\begin{aligned}
P(Y_1) &= P(A_1)P(A_2^c)P(A_3^c)P(A_4^c)P(A_5^c) = (0.1)(0.8)(0.7)(0.6)(0.5) = 0.0168,\\
P(Y_2) &= P(A_1^c)P(A_2)P(A_3^c)P(A_4^c)P(A_5^c) = (0.9)(0.2)(0.7)(0.6)(0.5) = 0.0378,\\
P(Y_3) &= P(A_1^c)P(A_2^c)P(A_3)P(A_4^c)P(A_5^c) = (0.9)(0.8)(0.3)(0.6)(0.5) = 0.0756,\\
P(Y_4) &= P(A_1^c)P(A_2^c)P(A_3^c)P(A_4)P(A_5^c) = (0.9)(0.8)(0.7)(0.4)(0.5) = 0.1008,\\
P(Y_5) &= P(A_1^c)P(A_2^c)P(A_3^c)P(A_4^c)P(A_5) = (0.9)(0.8)(0.7)(0.6)(0.5) = 0.1512.
\end{aligned}$$

Note that Y_i's are disjoint events. Hence we have

$$P(X_1) = \sum_{i=1}^{5} P(Y_i) = 0.0168 + 0.0378 + 0.0756 + 0.1008 + 0.1512 = 0.3816.$$

The probability that the error will be found by at least two tests is

$$\begin{aligned}
P(\text{by at least 2 tests}) &= 1 - P(\text{by 0 test}) - P(\text{by exactly 1 test})\\
&= 1 - P(X_0) - P(X_1)\\
&= 1 - 0.1512 - 0.3816\\
&= 0.4672.
\end{aligned}$$

So the probability that the error will be found by at least two tests is 46.72%.

(c) The probability that the error will be found by all five tests is

$$\begin{aligned}
P(\text{all 5 tests}) &= P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5)\\
&= (0.1)(0.2)(0.3)(0.4)(0.5)\\
&= 0.0012.
\end{aligned}$$

So the probability that the error will be found by all five tests is 0.12%.

2.12 A building is examined by policemen with four dogs that are trained to detect the scent of explosives. If there are explosives in a certain building, and each dog detects them with probability 0.6, independently of other dogs, what is the probability that the explosives will be detected by at least one dog?

Solution. Let D_i be the event that dog i detects the explosives. Then we have

$$\begin{aligned} P(\text{at least one dog detects}) &= 1 - P(\text{none of the dogs detect}) \\ &= 1 - P(D_1^c \cap D_2^c \cap D_3^c \cap D_4^c) \\ &= 1 - (1 - 0.6)^4 \\ &= 1 - (0.4)^4 \\ &= 1 - 0.0256 \\ &= 0.9744. \end{aligned}$$

So the probability that the explosives will be detected by at least one dog is 97.44%.

2.14 A spyware is trying to break into a system by guessing its password. It does not give up until it tries 1 million different passwords. What is the probability that it will guess the password and break in if by rules, the password must consist of

(a) 6 different lower-case letters?

(b) 6 different letters, some may be upper-case, and it is case-sensitive?

(c) any 6 letters, upper- or lower-case, and it is case-sensitive?

(d) any 6 characters including letters and digits?

Solution. We use $P(n, r)$ to denote the number of permutations of n items taken r at a time in this solution. ($P(\cdot)$ still denotes probability in other contexts.) Then we have

(a) There are 26 lower-case letters. The number of different passwords consisting of 6 different lower-case letters is $P(26, 6) = \frac{26!}{(26-6)!} = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21$. The probability that the spyware will guess the password in 1 million tries is $\frac{1{,}000{,}000}{P(26,6)}$.

(b) There are 52 letters (26 lower-case and 26 upper-case). The number of different passwords consisting of 6 different letters is $P(52, 6) = \frac{52!}{(52-6)!} = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47$. The probability that the spyware will guess the password in 1 million tries is $\frac{1{,}000{,}000}{P(52,6)}$.

(c) There are 52 letters (26 lower-case and 26 upper-case). The number of different passwords consisting of any 6 letters is $P_r(52, 6) = 52^6$ because replacement is allowed. The probability that the spyware will guess the password in 1 million tries is $\frac{1{,}000{,}000}{52^6}$.

(d) There are 62 characters (26 lower-case letters, 26 upper-case letters, and 10 digits). The number of different passwords consisting of any 6 characters is 62^6. The probability that the spyware will guess the password in 1 million tries is $\frac{1{,}000{,}000}{62^6}$.

2.15 A computer program consists of two blocks written independently by two different programmers. The first block has an error with probability 0.2. The second block has an error with probability 0.3. If the program returns an error, what is the probability that there is an error in both blocks?

Solution. Let A be the event that the first block has an error, and B be the event that the second block has an error. Then we have

$$\begin{aligned} P(A) &= 0.2, \\ P(B) &= 0.3, \\ P(A \cap B) &= P(A) \cdot P(B) = 0.2 \cdot 0.3 = 0.06. \end{aligned}$$

The probability that there is an error in both blocks given that the program returns an error is

$$P(A \cap B | A \cup B) = \frac{P(A \cap B)}{P(A \cup B)}.$$

Since $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.2 + 0.3 - 0.06 = 0.44$, we have

$$P(A \cap B | A \cup B) = \frac{0.06}{0.44} = \frac{3}{22} = 0.1364.$$

2.16 A computer maker receives parts from three suppliers, S1, S2, and S3. Fifty percent come from S1, twenty percent from S2, and thirty percent from S3. Among all the parts supplied by S1, 5% are defective. For S2 and S3, the portion of defective parts is 3% and 6%, respectively.

(a) What portion of all the parts is defective?

(b) A customer complains that a certain part in her recently purchased computer is defective. What is the probability that it was supplied by S1?

Solution.

(a) Let D be the event that a part is defective. Then we have

$$\begin{aligned} P(D|S_1) &= 0.05, \\ P(D|S_2) &= 0.03, \\ P(D|S_3) &= 0.06, \\ P(S_1) &= 0.5, \\ P(S_2) &= 0.2, \\ P(S_3) &= 0.3. \end{aligned}$$

The portion of all the parts that is defective is

$$\begin{aligned} P(D) &= P(D|S_1)P(S_1) + P(D|S_2)P(S_2) + P(D|S_3)P(S_3) \\ &= 0.05 \cdot 0.5 + 0.03 \cdot 0.2 + 0.06 \cdot 0.3 \\ &= 0.047. \end{aligned}$$

(b) The probability that a defective part was supplied by S1 is, by the Bayes' rule,

$$P(S_1|D) = \frac{P(D|S_1)P(S_1)}{P(D)} = \frac{0.05 \cdot 0.5}{0.047} = \frac{25}{47}.$$

2.18 A problem on a multiple-choice quiz is answered correctly with probability 0.9 if a student is prepared. An unprepared student guesses between 4 possible answers, so the probability of choosing the right answer is 1/4. Seventy-five percent of students prepare for the quiz. If

Mr. X gives a correct answer to this problem, what is the chance that he did not prepare for the quiz?

Solution. Let A be the event that a student is prepared, and C be the event that a student answers correctly. Then we have

$$\begin{aligned} P(C|A) &= 0.9, \\ P(C|A^c) &= 0.25, \\ P(A) &= 0.75, \\ P(A^c) &= 0.25. \end{aligned}$$

The probability that Mr. X did not prepare for the quiz given that he answered correctly is

$$P(A^c|C) = \frac{P(C|A^c)P(A^c)}{P(C)}.$$

To find $P(C)$, we use the law of total probability:

$$P(C) = P(C|A)P(A) + P(C|A^c)P(A^c) = 0.9 \cdot 0.75 + 0.25 \cdot 0.25 = 0.675 + 0.0625 = 0.7375.$$

Therefore,

$$P(A^c|C) = \frac{0.25 \cdot 0.25}{0.7375} = \frac{0.0625}{0.7375} = \frac{5}{59}.$$

2.19 At a plant, 20% of all the produced parts are subject to a special electronic inspection. It is known that any produced part which was inspected electronically has no defects with probability 0.95. For a part that was not inspected electronically this probability is only 0.7. A customer receives a part and find defects in it. What is the probability that this part went through an electronic inspection?

Solution. Let E be the event that a part was electronically inspected, and D be the event that a part has defects. Then we have

$$\begin{aligned} P(D|E) &= 0.05, \\ P(D|E^c) &= 0.3, \\ P(E) &= 0.2, \\ P(E^c) &= 0.8. \end{aligned}$$

The probability that a part with defects went through an electronic inspection is

$$P(E|D) = \frac{P(D|E)P(E)}{P(D)}.$$

To find $P(D)$, we use the law of total probability:

$$P(D) = P(D|E)P(E) + P(D|E^c)P(E^c) = 0.05 \cdot 0.2 + 0.3 \cdot 0.8 = 0.01 + 0.24 = 0.25.$$

Therefore,

$$P(E|D) = \frac{0.05 \cdot 0.2}{0.25} = \frac{0.01}{0.25} = \frac{1}{25}.$$

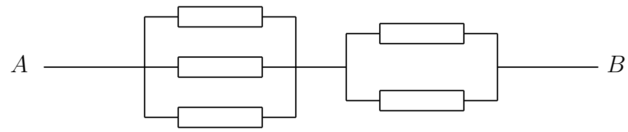

Figure 1: (Exercise 2.21) Calculate reliability of this system.

2.20 All athletes at the Olympic games are tested for performance-enhancing steroid drug use. The imperfect test gives positive results (indicating drug use) for 90% of all steroid-users but also (and incorrectly) for 2% of those who do not use steroids. Suppose that 5% of all registered athletes use steroids. If an athlete is tested negative, what is the probability that he/she uses steroids?

Solution. Let S be the event that an athlete uses steroids, and T be the event that an athlete tests positive. Then we have

$$\begin{aligned} P(T|S) &= 0.9, \\ P(T|S^c) &= 0.02, \\ P(S) &= 0.05, \\ P(S^c) &= 0.95. \end{aligned}$$

The probability that an athlete uses steroids given that he/she tests negative is

$$P(S|T^c) = \frac{P(T^c|S)P(S)}{P(T^c)}.$$

To find $P(T^c)$, we use the law of total probability:

$$P(T^c) = P(T^c|S)P(S) + P(T^c|S^c)P(S^c) = (1-0.9)\cdot 0.05 + (1-0.02)\cdot 0.95 = 0.936.$$

Therefore,

$$P(S|T^c) = \frac{(1-0.9)\cdot 0.05}{0.936} = \frac{0.005}{0.936} \approx 0.00534.$$

2.21 In the system in Figure 1, each component fails with probability 0.3 independently of other components. Compute the system's reliability.

Solution. Denote the three parallel components on the left as C_1, C_2, and C_3, and the two parallel components on the right as C_4 and C_5. The system works if at least one of C_1, C_2, or C_3 works and at least one of C_4 or C_5 works. The probability that at least one of C_1, C_2, or C_3 works is

$$P(C_1 \cup C_2 \cup C_3) = 1 - P(C_1^c \cap C_2^c \cap C_3^c) = 1 - (0.3)^3 = 1 - 0.027 = 0.973.$$

The probability that at least one of C_4 or C_5 works is

$$P(C_4 \cup C_5) = 1 - P(C_4^c \cap C_5^c) = 1 - (0.3)^2 = 1 - 0.09 = 0.91.$$

Since the two groups of components are independent, the system's reliability is

$$P(\text{system works}) = P(C_1 \cup C_2 \cup C_3) \cdot P(C_4 \cup C_5) = 0.973 \cdot 0.91 \approx 0.88543.$$

2.22 Three highways connect city A with city B. Two highways connect city B with city C. During a rush hour, each highway is blocked by a traffic accident with probability 0.2, independently of other highways.

(a) Compute the probability that there is at least one open route from A to C.

(b) How will a new highway, also blocked with probability 0.2 independently of other highways, change the probability in (a) if it is built

(i) between A and B?

(ii) between B and C?

(iii) between A and C?

Solution.

(a) Let H_{AB} be the event that there is at least one open highway from A to B, and H_{BC} be the event that there is at least one open highway from B to C. Then we have

$$P(H_{AB}) = 1 - P(\text{all highways from A to B are blocked}) = 1 - (0.2)^3 = 0.992,$$
$$P(H_{BC}) = 1 - P(\text{all highways from B to C are blocked}) = 1 - (0.2)^2 = 0.96.$$

The probability that there is at least one open route from A to C is

$$P(H_{AB} \cap H_{BC}) = P(H_{AB}) \cdot P(H_{BC}) = 0.992 \cdot 0.96 = 0.95232.$$

(b) (i) If a new highway is built between A and B, then there will be 4 highways from A to B. The probability that there is at least one open highway from A to B is

$$P(H_{AB}) = 1 - (0.2)^4 = 1 - 0.0016 = 0.9984.$$

The probability that there is at least one open route from A to C is

$$P(H_{AB} \cap H_{BC}) = P(H_{AB}) \cdot P(H_{BC}) = 0.9984 \cdot 0.96 = 0.958464.$$

(ii) If a new highway is built between B and C, then there will be 3 highways from B to C. The probability that there is at least one open highway from B to C is

$$P(H_{BC}) = 1 - (0.2)^3 = 1 - 0.008 = 0.992.$$

The probability that there is at least one open route from A to C is

$$P(H_{AB} \cap H_{BC}) = P(H_{AB}) \cdot P(H_{BC}) = 0.992 \cdot 0.992 = 0.984064.$$

(iii) If a new highway is built between A and C, then there will be a direct route from A to C. Let H_{AC} be the event that the direct route from A to C is open. Then the event that there is at least one open route from A to C is $(H_{AB} \cap H_{BC}) \cup H_{AC}$. Note that H_{AC} is independent of $H_{AB} \cap H_{BC}$. The probability that there is at least one open route from A to C is

$$\begin{aligned}
&P(\text{at least one open route from A to C}) \\
&= P((H_{AB} \cap H_{BC}) \cup H_{AC}) \\
&= P(H_{AB} \cap H_{BC}) + P(H_{AC}) - P((H_{AB} \cap H_{BC}) \cap H_{AC}) \\
&= P(H_{AB} \cap H_{BC}) + P(H_{AC}) - P(H_{AB} \cap H_{BC})P(H_{AC}) \\
&= 0.95232 + 0.8 - 0.95232 \cdot 0.8 \\
&= 0.95232 + 0.8 - 0.761856 \\
&= 0.990464.
\end{aligned}$$

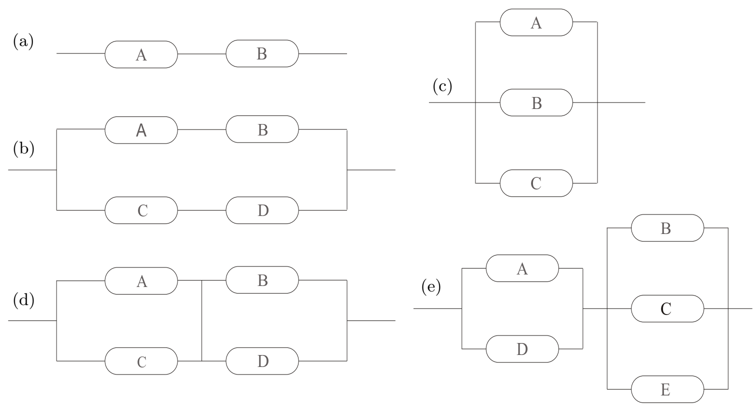

Figure 2: (Exercise 2.23) Calculate reliability of each system.

2.23 Calculate the reliability of each system shown in Figure 2, if components A, B, C, D, and E function properly with probabilities 0.9, 0.8, 0.7, 0.6, and 0.5, respectively.

Solution.

(a) The system works if both A and B work. The probability that the system works is

$$\begin{aligned} P(\text{system works}) &= P(A \cap B) \\ &= P(A)P(B) \\ &= 0.9 \cdot 0.8 \\ &= 0.72. \end{aligned}$$

(b) The system works if both A and B work, or both C and D work. The probability that the system works is

$$\begin{aligned} P(\text{system works}) &= P((A \cap B) \cup (C \cap D)) \\ &= P(A \cap B) + P(C \cap D) - P(A \cap B \cap C \cap D) \\ &= P(A)P(B) + P(C)P(D) - P(A)P(B)P(C)P(D) \\ &= 0.9 \cdot 0.8 + 0.7 \cdot 0.6 - 0.9 \cdot 0.8 \cdot 0.7 \cdot 0.6 \\ &= 0.72 + 0.42 - 0.3024 \\ &= 0.8376. \end{aligned}$$

(c) The system works if one of A, B, or C works. The probability that the system works is

$$\begin{aligned} P(\text{system works}) &= P(A \cup B \cup C) \\ &= 1 - P(A^c \cap B^c \cap C^c) \\ &= 1 - P(A^c)P(B^c)P(C^c) \\ &= 1 - (1 - 0.9)(1 - 0.8)(1 - 0.7) \\ &= 1 - (0.1)(0.2)(0.3) \\ &= 1 - 0.006 \\ &= 0.994. \end{aligned}$$

(d) The system works if at least one of A and C works, and meanwhile at least one of B and D works. The probability that the system works is

$$\begin{aligned}
P(\text{system works}) &= P((A \cup C) \cap (B \cup D)) \\
&= P(A \cup C)P(B \cup D) \\
&= (1 - P(A^c \cap C^c))(1 - P(B^c \cap D^c)) \\
&= (1 - P(A^c)P(C^c))(1 - P(B^c)P(D^c)) \\
&= (1 - (1 - 0.9)(1 - 0.7))(1 - (1 - 0.8)(1 - 0.6)) \\
&= (1 - (0.1)(0.3))(1 - (0.2)(0.4)) \\
&= (1 - 0.03)(1 - 0.08) \\
&= (0.97)(0.92) \\
&= 0.8924.
\end{aligned}$$

(e) The system works if at least one of A and D works, and meanwhile at least one of B, C, and E works. The probability that the system works is

$$\begin{aligned}
P(\text{system works}) &= P((A \cup D) \cap (B \cup C \cup E)) \\
&= P(A \cup D)P(B \cup C \cup E) \\
&= (1 - P(A^c \cap D^c))(1 - P(B^c \cap C^c \cap E^c)) \\
&= (1 - P(A^c)P(D^c))(1 - P(B^c)P(C^c)P(E^c)) \\
&= (1 - (1 - 0.9)(1 - 0.6))(1 - (1 - 0.8)(1 - 0.7)(1 - 0.5)) \\
&= (1 - (0.1)(0.4))(1 - (0.2)(0.3)(0.4)) \\
&= (1 - 0.04)(1 - 0.024) \\
&= (0.96)(0.976) \\
&= 0.93696.
\end{aligned}$$

2.24 Among 10 laptop computers, five are good and five have defects. Unaware of this, a customer buys 6 laptops.

(a) What is the probability of exactly 2 defective laptops among them?

(b) Given that at least 2 purchased laptops are defective, what is the probability that exactly 2 are defective?

Solution.

(a) The total number of ways to choose 6 laptops from 10 is $\binom{10}{6}$. The number of ways to choose exactly 2 defective laptops from 5 is $\binom{5}{2}$, and the number of ways to choose 4 good laptops from 5 is $\binom{5}{4}$. Therefore, the probability of exactly 2 defective laptops is

$$P(\text{exactly 2 defective}) = \frac{\binom{5}{2} \cdot \binom{5}{4}}{\binom{10}{6}} = \frac{10 \cdot 5}{210} = \frac{50}{210} = \frac{5}{21}.$$

(b) The probability of at least 2 defective laptops is

$$\begin{aligned}
P(\text{at least 2 defective}) &= 1 - P(\text{0 defective}) - P(\text{exactly 1 defective}) \\
&= 1 - \frac{\binom{5}{0}\cdot\binom{5}{6}}{\binom{10}{6}} - \frac{\binom{5}{1}\cdot\binom{5}{5}}{\binom{10}{6}} \\
&= 1 - 0 - \frac{5}{210} \\
&= \frac{41}{42}.
\end{aligned}$$

Therefore, the probability that exactly 2 are defective given that at least 2 purchased laptops are defective is

$$P(\text{exactly 2 defective}|\text{at least 2 defective}) = \frac{P(\text{exactly 2 defective})}{P(\text{at least 2 defective})} = \frac{\frac{5}{21}}{\frac{41}{42}} = \frac{10}{41}.$$

Note that we used the fact that

$$P(\text{exactly 2 defective} \cap \text{at least 2 defective}) = P(\text{exactly 2 defective})$$

since the event of exactly 2 defective laptops is a subset of the event of at least 2 defective laptops.

2.25 Two out of six computers in a lab have problems with hard drives. If three computers are selected at random for inspection, what is the probability that none of them has hard drive problems?

Solution. The total number of ways to choose 3 computers from 6 is $\binom{6}{3}$. The number of ways to choose 3 computers that do not have hard drive problems (i.e., from the 4 good computers) is $\binom{4}{3}$. Therefore, the probability that none of the selected computers has hard drive problems is

$$P(\text{none of the selected computers has hard drive problems}) = \frac{\binom{4}{3}}{\binom{6}{3}} = \frac{4}{20} = \frac{1}{5}.$$

2.26 For the University Cheese Club, Danielle and Anthony buy four brands of cheese, three brands of crackers, and two types of grapes. Each club member has to taste two different cheeses, two types of crackers, and one bunch of grapes. If 40 club members show up, can they all have different meals?

Solution. Notice that the number of different meals is equal to the product of the number of ways to choose 2 different cheeses from 4, the number of ways to choose 2 different crackers from 3, and the number of ways to choose 1 bunch of grapes from 2. We can calculate each of these as follows:

(i) The number of ways to choose 2 different cheeses from 4 is $\binom{4}{2} = 6$.

(ii) The number of ways to choose 2 different crackers from 3 is $\binom{3}{2} = 3$.

(iii) The number of ways to choose 1 bunch of grapes from 2 is $\binom{2}{1} = 2$.

Therefore, the total number of different meals is $6\cdot3\cdot2 = 36$. Since there are 40 club members, and only 36 different meals can be made, it is not possible for all 40 members to have different meals.

2.27 This is known as the Birthday Problem.

(a) Consider a class with 30 students. Compute the probability that at least two of them have their birthdays on the same day. (For simplicity, ignore the leap year.)

(b) How many students should be in class in order to have this probability above 0.5?

Solution.

(a) The probability that at least two students have the same birthday is equal to 1 minus the probability that all students have different birthdays. The total number of ways to assign birthdays to 30 students is 365^{30}. The number of ways to assign different birthdays to 30 students is $P(365, 30) = 365 \cdot 364 \cdot 363 \cdots (365 - 29) = \frac{365!}{(365-30)!}$. Therefore, the probability that all students have different birthdays is

$$P(\text{all different}) = \frac{\frac{365!}{(365-30)!}}{365^{30}}.$$

Thus, the probability that at least two students have the same birthday is

$$P(\text{at least two same}) = 1 - P(\text{all different}) = 1 - \frac{\frac{365!}{(365-30)!}}{365^{30}}.$$

Calculating this gives approximately 0.7063, or 70.63%.

(b) To find the number of students needed for the probability to be above 0.5, we can use the same formula and check different values of n until we find the smallest n such that

$$P(\text{at least two same}) = 1 - \frac{\frac{365!}{(365-n)!}}{365^{n}} > 0.5.$$

After calculations (e.g., write a program to compute this), it is found that with 23 students, the probability exceeds 0.5. Therefore, at least 23 students are needed in the class for the probability of at least two sharing a birthday to be above 0.5.

2.28 Among eighteen computers in some store, six have defects. Five randomly selected computers are bought for the university lab. Compute the probability that all five computers have no defects.

Solution. The total number of ways to choose 5 computers from 18 is $\binom{18}{5}$. The number of ways to choose 5 computers that have no defects (i.e., from the 12 good computers) is $\binom{12}{5}$. Therefore, the probability that all five selected computers have no defects is

$$P(\text{all five computers have no defects}) = \frac{\binom{12}{5}}{\binom{18}{5}} = \frac{11}{119}.$$

2.29 A quiz consists of 6 multiple-choice questions. Each question has 4 possible answers. A student is unprepared, and he has no choice but guessing answers completely at random. He passes the quiz if he gets at least 3 questions correctly. What is the probability that he will pass?

Solution. Let X be the number of questions answered correctly. Then X follows a binomial distribution with parameters $n = 6$ and $p = \frac{1}{4}$. The probability that the student passes the quiz is

$$P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6).$$

Calculating each term:

$$P(X=3) = \binom{6}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^3 = 20\cdot\frac{1}{64}\cdot\frac{27}{64} = \frac{540}{4096},$$

$$P(X=4) = \binom{6}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^2 = 15\cdot\frac{1}{256}\cdot\frac{9}{16} = \frac{135}{4096},$$

$$P(X=5) = \binom{6}{5}\left(\frac{1}{4}\right)^5\left(\frac{3}{4}\right)^1 = 6\cdot\frac{1}{1024}\cdot\frac{3}{4} = \frac{18}{4096},$$

$$P(X=6) = \binom{6}{6}\left(\frac{1}{4}\right)^6\left(\frac{3}{4}\right)^0 = 1\cdot\frac{1}{4096}\cdot 1 = \frac{1}{4096}.$$

Adding these probabilities together:

$$P(X\geq 3) = \frac{540+135+18+1}{4096} = \frac{694}{4096} \approx 0.1694.$$

2.30 An internet search engine looks for a keyword in 9 databases, searching them in a random order. Only 5 of these databases contain the given keyword. Find the probability that it will be found in at least 2 of the first 4 searched databases.

Solution. Let X be the number of databases containing the keyword that are searched. The probability that it will be found in at least 2 of the first 4 searched databases is

$$P(X\geq 2) = P(X=2)+P(X=3)+P(X=4).$$

For $X=2$, we need to choose 2 databases containing the keyword from the 5 that have it, and 2 databases not containing the keyword from the 4 that do not have it. The total number of ways to choose 4 databases from 9 is $\binom{9}{4}$. Therefore,

$$P(X=2) = \frac{\binom{5}{2}\binom{4}{2}}{\binom{9}{4}}.$$

For $X=3$, we need to choose 3 databases containing the keyword from the 5 that have it, and 1 database not containing the keyword from the 4 that do not have it. Therefore,

$$P(X=3) = \frac{\binom{5}{3}\binom{4}{1}}{\binom{9}{4}}.$$

For $X=4$, we need to choose 4 databases containing the keyword from the 5 that have it, and 0 databases not containing the keyword from the 4 that do not have it. Therefore,

$$P(X=4) = \frac{\binom{5}{4}\binom{4}{0}}{\binom{9}{4}}.$$

Adding these probabilities together:

$$P(X\geq 2) = \frac{60+40+5}{126} = \frac{105}{126} = \frac{5}{6}.$$

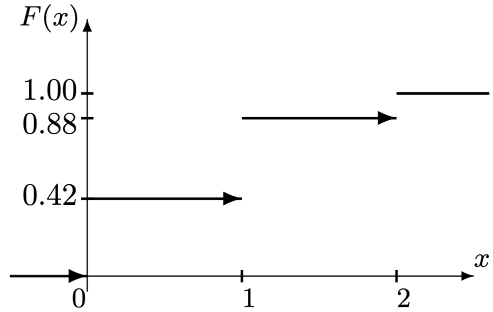

Figure 3: The cdf F of Exercise 3.1(b) solution.

Chapter 3

3.1 A computer virus is trying to corrupt two files. The first file will be corrupted with probability 0.4. Independently of it, the second file will be corrupted with probability 0.3.

(a) Compute the probability mass function (pmf) of X, the number of corrupted files.

(b) Draw a graph of its cumulative distribution function (cdf).

Solution.

(a) Let X be the number of corrupted files. The possible values of X are 0, 1, and 2. We can compute the probabilities as follows:

$$\begin{aligned}
P(X=0) &= P(\text{first file not corrupted}) \cdot P(\text{second file not corrupted}) \\
&= (1-0.4)(1-0.3) = 0.6 \cdot 0.7 = 0.42, \\
P(X=1) &= P(\text{first file corrupted, second file not corrupted}) \\
&\quad + P(\text{first file not corrupted, second file corrupted}) \\
&= 0.4 \cdot 0.7 + 0.6 \cdot 0.3 = 0.28 + 0.18 = 0.46, \\
P(X=2) &= P(\text{first file corrupted}) \cdot P(\text{second file corrupted}) = 0.4 \cdot 0.3 = 0.12.
\end{aligned}$$

Therefore, the pmf of X is:

x	0	1	2
$P(X=x)$	0.42	0.46	0.12

(b) The cumulative distribution function (cdf) of X is defined as $F(x) = P(X \leq x)$. We can compute the cdf values as follows:

$$\begin{aligned}
F(0) &= P(X \leq 0) = P(X=0) = 0.42, \\
F(1) &= P(X \leq 1) = P(X=0) + P(X=1) = 0.42 + 0.46 = 0.88, \\
F(2) &= P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = 0.42 + 0.46 + 0.12 = 1.
\end{aligned}$$

The cdf F can be represented graphically (see Figure 3) as a step function with jumps. Note that F is right-continuous but not left-continuous at $x=0$, $x=1$, and $x=2$.

3.2 Every day, the number of network blackouts has a distribution (probability mass function)

x	0	1	2
$P(x)$	0.7	0.2	0.1

A small internet trading company estimates that each network blackout results in a \$500 loss. Compute expectation and variance of this company's daily loss due to blackouts.

Solution. The expected daily loss L can be computed as follows:

$$\mathbb{E}(L) = \sum_{x=0}^{2} P(x) \cdot (\text{loss per blackout} \cdot x) = 0.7 \cdot 0 + 0.2 \cdot 500 + 0.1 \cdot 1000 = 0 + 100 + 100 = 200.$$

The variance of the daily loss can be computed as follows:

$$\begin{aligned}
\text{Var}(L) &= \mathbb{E}(L^2) - (\mathbb{E}(L))^2 \\
&= \sum_{x=0}^{2} P(x) \cdot (\text{loss per blackout} \cdot x)^2 - 200^2 \\
&= 0.7 \cdot 0 + 0.2 \cdot 250000 + 0.1 \cdot 1000000 - 40000 \\
&= 0 + 50000 + 100000 - 40000 \\
&= 110000.
\end{aligned}$$

3.3 There is one error in one of five blocks of a program. To find the error, we test three randomly selected blocks. Let X be the number of errors in these three blocks. Compute $\mathbb{E}(X)$ and $\text{Var}(X)$.

Solution. The expected number of errors in the three randomly selected blocks can be computed as follows:

$$\mathbb{E}(X) = \sum_{x=0}^{1} x \cdot P(X = x) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) = 0 \cdot \frac{\binom{4}{3}}{\binom{5}{3}} + 1 \cdot \frac{\binom{1}{1} \cdot \binom{4}{2}}{\binom{5}{3}} = 0 + 1 \cdot \frac{6}{10} = 0.6.$$

The variance can be computed as follows:

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_{x=0}^{1} x^2 \cdot P(X = x) - (0.6)^2 = 0^2 \cdot \frac{4}{10} + 1^2 \cdot \frac{6}{10} - 0.36 = 0.24.$$

3.5 A software package consists of 12 programs, five of which must be upgraded. If 4 programs are randomly chosen for testing,

(a) What is the probability that at least two of them must be upgraded?

(b) What is the expected number of programs, out of the chosen four, that must be upgraded?

Solution.

(a) The probability that at least two of the chosen programs must be upgraded can be computed as follows:

$$P(\text{at least 2 upgraded}) = 1 - P(0 \text{ upgraded}) - P(1 \text{ upgraded}).$$

The total number of ways to choose 4 programs from 12 is $\binom{12}{4}$. The number of ways to choose 0 upgraded programs from 5 is $\binom{5}{0}$, and the number of ways to choose 4 non-upgraded programs from 7 is $\binom{7}{4}$. Therefore,

$$P(0 \text{ upgraded}) = \frac{\binom{5}{0} \cdot \binom{7}{4}}{\binom{12}{4}} = \frac{1 \cdot 35}{495} = \frac{35}{495}.$$

The number of ways to choose 1 upgraded program from 5 is $\binom{5}{1}$, and the number of ways to choose 3 non-upgraded programs from 7 is $\binom{7}{3}$. Therefore,

$$P(1 \text{ upgraded}) = \frac{\binom{5}{1} \cdot \binom{7}{3}}{\binom{12}{4}} = \frac{5 \cdot 35}{495} = \frac{175}{495}.$$

Thus,

$$P(\text{at least 2 upgraded}) = 1 - \frac{35}{495} - \frac{175}{495} = 1 - \frac{210}{495} = \frac{285}{495}.$$

(b) The expected number of programs that must be upgraded can be computed as follows:

$$\mathbb{E}(X) = \sum_{x=0}^{4} x \cdot P(X = x).$$

Since X can only take values 0, 1, 2, 3, or 4, we have:

$$\mathbb{E}(X) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4).$$

We already computed $P(X = 0)$ and $P(X = 1)$. Similarly, we can compute $P(X = 2)$, $P(X = 3)$, and $P(X = 4)$. There are

$$P(X = 2) = \frac{\binom{5}{2} \cdot \binom{7}{2}}{\binom{12}{4}} = \frac{10 \cdot 21}{495} = \frac{210}{495},$$
$$P(X = 3) = \frac{\binom{5}{3} \cdot \binom{7}{1}}{\binom{12}{4}} = \frac{10 \cdot 7}{495} = \frac{70}{495},$$
$$P(X = 4) = \frac{\binom{5}{4} \cdot \binom{7}{0}}{\binom{12}{4}} = \frac{5 \cdot 1}{495} = \frac{5}{495}.$$

Adding these probabilities together, we have:

$$\mathbb{E}(X) = 0 \cdot \frac{35}{495} + 1 \cdot \frac{175}{495} + 2 \cdot \frac{210}{495} + 3 \cdot \frac{70}{495} + 4 \cdot \frac{5}{495} = \frac{0 + 175 + 420 + 210 + 20}{495} = \frac{825}{495} = \frac{11}{6}.$$

3.6 A computer program contains one error. In order to find the error, we split the program into 6 blocks and test two of them, selected at random. Let X be the number of errors in these blocks. Compute $\mathbb{E}(X)$.

Solution. The expected number of errors in the two randomly selected blocks can be computed as follows:

$$\mathbb{E}(X) = \sum_{x=0}^{1} x \cdot P(X = x) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) = 0 \cdot \frac{\binom{5}{2}}{\binom{6}{2}} + 1 \cdot \frac{\binom{1}{1} \cdot \binom{5}{1}}{\binom{6}{2}} = 0 + 1 \cdot \frac{5}{15} = \frac{1}{3}.$$

3.7 The number of home runs scored by a certain team in one baseball game is a random variable with the distribution

x	0	1	2
$P(x)$	0.4	0.4	0.2

The team plays 2 games. The number of home runs scored in one game is independent of the number of home runs in the other game. Let Y be the total number of home runs. Find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$.

Solution. The expected number of home runs in one game is

$$\mathbb{E}(X) = 0 \cdot 0.4 + 1 \cdot 0.4 + 2 \cdot 0.2 = 0 + 0.4 + 0.4 = 0.8.$$

Since the team plays 2 games, the expected total number of home runs is

$$\mathbb{E}(Y) = 2 \cdot \mathbb{E}(X) = 2 \cdot 0.8 = 1.6.$$

The variance of the number of home runs in one game is

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = (0^2 \cdot 0.4 + 1^2 \cdot 0.4 + 2^2 \cdot 0.2) - (0.8)^2 = (0 + 0.4 + 0.8) - 0.64 = 0.56.$$

Since the number of home runs in one game is independent of the number of home runs in the other game, the variance of the total number of home runs is

$$\mathrm{Var}(Y) = 2 \cdot \mathrm{Var}(X) = 2 \cdot 0.56 = 1.12.$$

3.8 A computer user tries to recall her password. She knows it can be one of 4 possible passwords. She tries her passwords until she finds the right one. Let X be the number of wrong passwords she uses before she finds the right one. Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

Solution. Let Y be the total number of tries to find the right one. Then Y follows a geometric distribution with parameter $p = \frac{1}{4}$. The expected value and variance of a geometric distribution are given by

$$\mathbb{E}(Y) = \frac{1}{p} = \frac{1}{\frac{1}{4}} = 4,$$

and

$$\mathrm{Var}(Y) = \frac{1-p}{p^2} = \frac{1-\frac{1}{4}}{\left(\frac{1}{4}\right)^2} = \frac{\frac{3}{4}}{\frac{1}{16}} = 12.$$

Since $X = Y - 1$, we have

$$\mathbb{E}(X) = \mathbb{E}(Y) - 1 = 4 - 1 = 3,$$

and

$$\mathrm{Var}(X) = \mathrm{Var}(Y) = 12.$$

3.20 A quality control engineer tests the quality of produced computers. Suppose that 5% of computers have defects, and defects occur independently of each other.

(a) Find the probability of exactly 3 defective computers in a shipment of twenty.

(b) Find the probability that the engineer has to test at least 5 computers in order to find 2 defective ones.

Solution.

(a) Let X be the number of defective computers in a shipment of 20. Then X follows a binomial distribution with parameters $n = 20$ and $p = 0.05$. The probability of exactly 3 defective computers is

$$P(X = 3) = \binom{20}{3}(0.05)^3(0.95)^{17} = 1140 \cdot 0.000125 \cdot 0.4228 \approx 0.0605.$$

(b) Let Y be the number of computers tested until 2 defective ones are found. Then Y follows a negative binomial distribution with parameters $r = 2$ and $p = 0.05$. The probability that the engineer has to test at least 5 computers in order to find 2 defective ones is

$$P(Y \geq 5) = 1 - P(Y < 5) = 1 - P(Y = 2) - P(Y = 3) - P(Y = 4).$$

Recall that the pmf of a negative binomial distribution is given by

$$P(Y = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \ldots.$$

Calculating each term above:

$$P(Y = 2) = \binom{1}{1}(0.05)^2(0.95)^0 = 1 \cdot 0.0025 \cdot 1 = 0.0025,$$

$$P(Y = 3) = \binom{2}{1}(0.05)^2(0.95)^1 = 2 \cdot 0.0025 \cdot 0.95 = 0.00475,$$

$$P(Y = 4) = \binom{3}{1}(0.05)^2(0.95)^2 = 3 \cdot 0.0025 \cdot 0.9025 \approx 0.00677.$$

Adding these probabilities together:

$$P(Y < 5) = 0.0025 + 0.00475 + 0.00677 \approx 0.01402.$$

Therefore,

$$P(Y \geq 5) = 1 - 0.01402 \approx 0.98598.$$

(Hint: Another way to solve this problem is letting Z be the number of successes in 4 trials, and then $P(Y \geq 5) = P(Z < 2) = P(Z = 0) + P(Z = 1)$, where Z follows a binomial distribution with parameters $n = 4$ and $p = 0.05$.)

3.21 A lab network consisting of 20 computers was attacked by a computer virus. This virus enters each computer with probability 0.4, independently of other computers. Find the probability that it entered at least 10 computers.

Solution. Let X be the number of computers that the virus entered. Then X follows a binomial distribution with parameters $n = 20$ and $p = 0.4$. The probability that the virus entered at least 10 computers is

$$P(X \geq 10) = 1 - P(X < 10) = 1 - \sum_{k=0}^{9} P(X = k) = 1 - \sum_{k=0}^{9} \binom{20}{k}(0.4)^k(0.6)^{20-k} \approx 0.2447.$$

3.22 Five percent of computer parts produced by a certain supplier are defective. What is the probability that a sample of 16 parts contains more than 3 defective ones?

Solution. Let X be the number of defective parts in a sample of 16. Then X follows a binomial distribution with parameters $n = 16$ and $p = 0.05$. The probability that the sample contains more than 3 defective parts is

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \sum_{k=0}^{3} P(X = k) = 1 - \sum_{k=0}^{3} \binom{16}{k} (0.05)^k (0.95)^{16-k} \approx 0.0070.$$

3.23 Every day, a lecture may be canceled due to inclement weather with probability 0.05. Class cancellations on different days are independent.

(a) There are 15 classes left this semester. Compute the probability that at least 4 of them get canceled.

(b) Compute the probability that the tenth class this semester is the third class that gets canceled.

Solution.

(a) Let X be the number of classes canceled in 15 days. Then X follows a binomial distribution with parameters $n = 15$ and $p = 0.05$. The probability that at least 4 classes get canceled is

$$P(X \geq 4) = 1 - P(X < 4) = 1 - \sum_{k=0}^{3} P(X = k) = 1 - \sum_{k=0}^{3} \binom{15}{k} (0.05)^k (0.95)^{15-k}.$$

We can write a program to compute this sum. Or we know $P(X < 4) = P(X \leq 3) = F(3)$, where F is the cdf of a binomial distribution with parameters $n = 15$ and $p = 0.05$. From Table A2 of the textbook, we find that $F(3) \approx 0.995$. Therefore,

$$P(X \geq 4) \approx 1 - 0.995 = 0.005.$$

(b) Let Y be the number of classes canceled before the tenth class. We want to find the probability that the tenth class is the third class that gets canceled. This means that in the first 9 classes, exactly 2 classes were canceled, and the tenth class is canceled. The probability of this event is

$$\begin{aligned} P(Y = 2 \text{ and tenth class canceled}) &= P(Y = 2) \cdot P(\text{tenth class canceled}) \\ &= \binom{9}{2} (0.05)^2 (0.95)^7 \cdot 0.05 \approx 0.0031. \end{aligned}$$

Note that we used the fact that the probability of the tenth class being canceled is independent of the first 9 classes to get the first equality.

3.24 An internet search engine looks for a certain keyword in a sequence of independent web sites. It is believed that 20% of the sites contain this keyword.

(a) Compute the probability that at least 5 of the first 10 sites contain the given keyword.

(b) Compute the probability that the search engine had to visit at least 5 sites in order to find the first occurrence of a keyword.

Solution.

(a) Let X be the number of sites containing the keyword in the first 10 sites. Then X follows a binomial distribution with parameters $n = 10$ and $p = 0.2$. The probability that at least 5 of the first 10 sites contain the keyword is

$$P(X \geq 5) = 1 - P(X < 5) = 1 - \sum_{k=0}^{4} P(X = k) = 1 - \sum_{k=0}^{4} \binom{10}{k} (0.2)^k (0.8)^{10-k}.$$

We can write a program to compute this sum. Or we know $P(X < 5) = P(X \leq 4) = F(4)$, where F is the cdf of a binomial distribution with parameters $n = 10$ and $p = 0.2$. Using statistical software or a calculator, we find that $F(4) \approx 0.9672$. Therefore,

$$P(X \geq 5) \approx 1 - 0.9672 = 0.0328.$$

(b) Let Y be the number of sites visited until the first occurrence of the keyword. Then Y follows a geometric distribution with parameter $p = 0.2$. The probability that the search engine had to visit at least 5 sites in order to find the first occurrence of a keyword is

$$P(Y \geq 5) = \sum_{k=5}^{\infty} (1-p)^{k-1} p = (1-p)^4 p \frac{1}{1-(1-p)} = (0.8)^4 \cdot 0.2 \cdot \frac{1}{0.2} = (0.8)^4 \approx 0.4096.$$

3.25 About ten percent of users do not close Windows properly. Suppose that Windows is installed in a public library that is used by random people in a random order.

(a) On the average, how many users of this computer do not close Windows properly before someone does close it properly?

(b) What is the probability that exactly 8 of the next 10 users will close Windows properly?

Solution.

(a) Let X be the number of users who do not close Windows properly before someone does close it properly. Then $Z = X + 1$ is the total number of users checked until the first proper closure, and Z follows a geometric distribution with parameter $p = 0.9$. The expected value of a geometric distribution is given by

$$\mathbb{E}(Z) = \frac{1}{p} = \frac{1}{0.9} \approx 1.1111.$$

Hence the expected number of users who do not close Windows properly before someone does close it properly is

$$\mathbb{E}(X) = \mathbb{E}(Z) - 1 \approx 1.1111 - 1 = 0.1111.$$

(b) Let Y be the number of users who close Windows properly in the next 10 users. Then Y follows a binomial distribution with parameters $n = 10$ and $p = 0.9$. The probability that exactly 8 of the next 10 users will close Windows properly is

$$P(Y = 8) = \binom{10}{8} (0.9)^8 (0.1)^2 = 45 \cdot 0.43046721 \cdot 0.01 \approx 0.1937.$$

3.26 After a computer virus entered the system, a computer manager checks the condition of all important files. She knows that each file has probability 0.2 to be damaged by the virus, independently of other files.

(a) Compute the probability that at least 5 of the first 20 files are damaged.

(b) Compute the probability that the manager has to check at least 6 files in order to find 3 undamaged files.

Solution.

(a) Let X be the number of damaged files in the first 20 files. Then X follows a binomial distribution with parameters $n = 20$ and $p = 0.2$. The probability that at least 5 of the first 20 files are damaged is

$$P(X \geq 5) = 1 - P(X < 5) = 1 - \sum_{k=0}^{4} P(X = k) = 1 - \sum_{k=0}^{4} \binom{20}{k} (0.2)^k (0.8)^{20-k} \approx 0.37.$$

(b) Let Y be the number of files checked until 3 undamaged files are found. Then Y follows a negative binomial distribution with parameters $r = 3$ and $p = 0.8$. The probability that the manager has to check at least 6 files in order to find 3 undamaged files is

$$P(Y \geq 6) = 1 - P(Y < 6) = 1 - \sum_{k=3}^{5} P(Y = k) = 1 - \sum_{k=3}^{5} \binom{k-1}{2} (0.8)^3 (0.2)^{k-3}.$$

3.28 Messages arrive at an electronic message center at random times, with an average of 9 messages per hour.

(a) What is the probability of receiving at least five messages during the next hour?

(b) What is the probability of receiving exactly five messages during the next hour?

Solution.

(a) Let X be the number of messages received during the next hour. Then X follows a Poisson distribution with parameter $\lambda = 9$. The probability of receiving at least five messages is

$$P(X \geq 5) = 1 - P(X < 5) = 1 - \sum_{k=0}^{4} P(X = k) = 1 - \sum_{k=0}^{4} \frac{e^{-9} 9^k}{k!}.$$

We can use Table A3 of the textbook to find that $P(X < 5) = F(4) = 0.055$ where F is the cdf of a Poisson distribution with parameter $\lambda = 9$. Therefore,

$$P(X \geq 5) \approx 1 - 0.055 = 0.945.$$

(b) The probability of receiving exactly five messages is

$$P(X = 5) = \frac{e^{-9} 9^5}{5!}.$$

Similarly, we can use Table A3 of the textbook to find that $P(X = 5) = F(5) - F(4) = 0.116 - 0.055 = 0.061$.

3.30 An insurance company divides its customers into 2 groups. Twenty percent of customers are in the high-risk group, and eighty percent are in the low-risk group. The high-risk customers make an average of 1 accident per year while the low-risk customers make an average of 0.1 accidents per year. Eric had no accidents last year. What is the probability that he is a high-risk driver?

Solution. Let H be the event that Eric is a high-risk driver, and let X be the number of accidents that Eric had last year. We want to find $P(H|X = 0)$. By Bayes' rule, we have

$$P(H|X = 0) = \frac{P(X = 0|H)P(H)}{P(X = 0)}.$$

We know that $P(H) = 0.2$ and $P(L) = 0.8$, where L is the event that Eric is a low-risk driver. Given that Eric is a high-risk driver, the number of accidents he has in a year follows a Poisson distribution with parameter $\lambda = 1$. Therefore, the probability of having no accidents given that Eric is a high-risk driver is

$$P(X = 0|H) = \frac{e^{-1}1^0}{0!} = e^{-1} \approx 0.3679.$$

Given that Eric is a low-risk driver, the number of accidents he has in a year follows a Poisson distribution with parameter $\lambda = 0.1$. Therefore, the probability of having no accidents given that Eric is a low-risk driver is

$$P(X = 0|L) = \frac{e^{-0.1}0.1^0}{0!} = e^{-0.1} \approx 0.9048.$$

The total probability of having no accidents is

$$P(X = 0) = P(X = 0|H)P(H) + P(X = 0|L)P(L) = 0.3679 \cdot 0.2 + 0.9048 \cdot 0.8 \approx 0.79742.$$

Therefore, the conditional probability that Eric is a high-risk driver given that he had no accidents last year is

$$P(H|X = 0) = \frac{P(X = 0|H)P(H)}{P(X = 0)} = \frac{0.3679 \cdot 0.2}{0.79742} \approx \frac{0.07358}{0.79742} \approx 0.0923.$$

3.31 Eric from Exercise 3.30 continues driving. After three years, he still has no traffic accidents. Now, what is the conditional probability that he is a high-risk driver?

Solution. The high-risk customers make an average of 1 accident per year while the low-risk customers make an average of 0.1 accidents per year. Therefore, after three years, the number of accidents for high-risk customers follows a Poisson distribution with parameter $\lambda = 3$, and the number of accidents for low-risk customers follows a Poisson distribution with parameter $\lambda = 0.3$.

Let H be the event that Eric is a high-risk driver, and let Y be the number of accidents Eric had in three years. We want to find $P(H|Y = 0)$. By Bayes' rule, we have

$$P(H|Y = 0) = \frac{P(Y = 0|H)P(H)}{P(Y = 0)}.$$

We know that $P(H) = 0.2$ and $P(L) = 0.8$, where L is the event that Eric is a low-risk driver. Given that Eric is a high-risk driver, the probability of having no accidents in three years is

$$P(Y = 0|H) = \frac{e^{-3}3^0}{0!} = e^{-3} \approx 0.0498.$$

Given that Eric is a low-risk driver, the probability of having no accidents in three years is

$$P(Y = 0|L) = \frac{e^{-0.3}0.3^0}{0!} = e^{-0.3} \approx 0.7408.$$

The total probability of having no accidents in three years is

$$P(Y = 0) = P(Y = 0|H)P(H) + P(Y = 0|L)P(L) = 0.0498 \cdot 0.2 + 0.7408 \cdot 0.8 \approx 0.593.$$

Therefore, the conditional probability that Eric is a high-risk driver given that he had no accidents in three years is

$$P(H|Y = 0) = \frac{P(Y = 0|H)P(H)}{P(Y = 0)} = \frac{0.0498 \cdot 0.2}{0.593} \approx \frac{0.00996}{0.593} \approx 0.0168.$$

3.32 Before the computer is assembled, its vital component (motherboard) goes through a special inspection. Only 80% of components pass this inspection.

(a) What is the probability that at least 18 of the next 20 components pass inspection?

(b) On the average, how many components should be inspected until a component that passes inspection is found?

Solution.

(a) Let X be the number of components that pass inspection in the next 20 components. Then X follows a binomial distribution with parameters $n = 20$ and $p = 0.8$. The probability that at least 18 of the next 20 components pass inspection is

$$\begin{aligned} P(X \geq 18) &= P(X = 18) + P(X = 19) + P(X = 20) \\ &= \binom{20}{18}(0.8)^{18}(0.2)^2 + \binom{20}{19}(0.8)^{19}(0.2)^1 + \binom{20}{20}(0.8)^{20}(0.2)^0 \\ &= \binom{20}{2}(0.8)^{18}(0.2)^2 + \binom{20}{1}(0.8)^{19}(0.2)^1 + \binom{20}{0} \cdot (0.8)^{20}(0.2)^0 \\ &= F(2) \end{aligned}$$

where F is the cumulative distribution function of the binomial distribution with parameters $n = 20$ and $p = 0.2$ (not $p = 0.8$!). From Table A2, we find that $F(2) \approx 0.206$. Therefore, the probability that at least 18 of the next 20 components pass inspection is approximately 0.206.

(b) Let Y be the number of components inspected until a component that passes inspection is found. Then Y follows a geometric distribution with parameter $p = 0.8$. The expected value of a geometric distribution is given by

$$\mathbb{E}(Y) = \frac{1}{p} = \frac{1}{0.8} = 1.25.$$

3.33 On the average, 1 computer in 800 crashes during a severe thunderstorm. A certain company had 4,000 working computers when the area was hit by a severe thunderstorm.

(a) Compute the probability that less than 10 computers crashed.

(b) Compute the probability that exactly 10 computers crashed.

Solution.

(a) Let X be the number of computers that crashed during the severe thunderstorm. Then X follows a binomial distribution with parameters $n = 4000$ and $p = \frac{1}{800} = 0.00125$. The probability that less than 10 computers crashed is

$$P(X < 10) = \sum_{k=0}^{9} P(X = k) = \sum_{k=0}^{9} \binom{4000}{k} (0.00125)^k (0.99875)^{4000-k}.$$

Let $\lambda = np = 4000 \cdot 0.00125 = 5$. Since n is large and p is small, we can approximate the binomial distribution with a Poisson distribution with parameter $\lambda = 5$. Therefore, we have

$$P(X < 10) \approx \sum_{k=0}^{9} \frac{e^{-5} 5^k}{k!} = F(9),$$

where F is the cumulative distribution function of the Poisson distribution with parameter $\lambda = 5$. From Table A3, we find that $F(9) \approx 0.968$. Therefore, the probability that less than 10 computers crashed is approximately 0.968.

(b) The probability that exactly 10 computers crashed is

$$P(X = 10) = \binom{4000}{10} (0.00125)^{10} (0.99875)^{3990}.$$

Let $\lambda = np = 4000 \cdot 0.00125 = 5$. Since n is large and p is small, we can approximate the binomial distribution with a Poisson distribution with parameter $\lambda = 5$. Therefore, we have

$$P(X = 10) \approx \frac{e^{-5} 5^{10}}{10!} = F(10) - F(9),$$

where F is the cumulative distribution function of the Poisson distribution with parameter $\lambda = 5$. From Table A3, we find that $F(10) \approx 0.986$ and $F(9) \approx 0.968$. Therefore, the probability that exactly 10 computers crashed is approximately $0.986 - 0.968 = 0.018$.

3.34 The number of computer shutdowns during any month has a Poisson distribution, averaging 0.25 shutdowns per month.

(a) What is the probability of at least 3 computer shutdowns during the next year?

(b) During the next year, what is the probability of at least 3 months (out of 12) with exactly 1 computer shutdown in each?

Solution.

(a) Let X be the number of computer shutdowns during the next year. Then X follows a Poisson distribution with parameter $\lambda = 0.25 \times 12 = 3$. The probability of at least 3 computer shutdowns is

$$\begin{aligned} P(X \geq 3) = 1 - P(X < 3) = 1 - \sum_{k=0}^{2} P(X = k) = 1 - \sum_{k=0}^{2} \frac{e^{-3} 3^k}{k!} \\ = 1 - F(2) = 1 - 0.423 = 0.577. \end{aligned}$$

Here F is the cumulative distribution function of the Poisson distribution with parameter $\lambda = 3$. From Table A3, we find that $F(2) \approx 0.423$. Therefore, the probability of at least 3 computer shutdowns during the next year is approximately 0.577.

(b) Let Y be the number of months with exactly 1 computer shutdown during the next year. Then Y follows a binomial distribution with parameters $n = 12$ and $p = P(X = 1)$, where X is the number of computer shutdowns in a month. The probability of exactly 1 computer shutdown in a month is

$$P(X = 1) = \frac{e^{-0.25}0.25^1}{1!} = 0.25e^{-0.25}.$$

The probability of at least 3 months with exactly 1 computer shutdown is

$$P(Y \geq 3) = 1 - \sum_{k=0}^{2} P(Y = k) = 1 - \sum_{k=0}^{2} \binom{12}{k} (0.25e^{-0.25})^k (1 - 0.25e^{-0.25})^{12-k}.$$

3.35 A dangerous computer virus attacks a folder consisting of 250 files. Files are affected by the virus independently of one another. Each file is affected with the probability 0.032. What is the probability that more than 7 files are affected by this virus?

Solution. Let X be the number of files affected by the virus. Then X follows a binomial distribution with parameters $n = 250$ and $p = 0.032$. The probability that more than 7 files are affected is

$$P(X > 7) = 1 - P(X \leq 7) = 1 - \sum_{k=0}^{7} P(X = k) = 1 - \sum_{k=0}^{7} \binom{250}{k} (0.032)^k (0.968)^{250-k}.$$

Let $\lambda = np = 250 \cdot 0.032 = 8$. Since n is large and p is small, we can approximate the binomial distribution with a Poisson distribution with parameter $\lambda = 8$. Therefore, we have

$$P(X > 7) \approx 1 - \sum_{k=0}^{7} \frac{e^{-8}8^k}{k!} = 1 - F(7),$$

where F is the cumulative distribution function of the Poisson distribution with parameter $\lambda = 8$. From Table A3, we find that $F(7) \approx 0.453$. Therefore, the probability that more than 7 files are affected by this virus is approximately $1 - 0.453 = 0.547$.

3.36 In some city, the probability of a thunderstorm on any day is 0.6. During a thunderstorm, the number of traffic accidents has Poisson distribution with parameter 10. Otherwise, the number of traffic accidents has Poisson distribution with parameter 4. If there were 7 accidents yesterday, what is the probability that there was a thunderstorm?

Solution. Let T be the event that there was a thunderstorm yesterday, and let A be the event that there were 7 accidents yesterday. We want to find $P(T|A)$. By Bayes' rule, we have

$$P(T|A) = \frac{P(A|T)P(T)}{P(A)}.$$

We know that $P(T) = 0.6$ and $P(\bar{T}) = 0.4$, where $\bar{T}$ is the event that there was no thunderstorm yesterday. The probability of having 7 accidents given that there was a thunderstorm is

$$P(A|T) = \frac{e^{-10}10^7}{7!},$$

and the probability of having 7 accidents given that there was no thunderstorm is

$$P(A|\bar{T}) = \frac{e^{-4}4^7}{7!}.$$

The total probability of having 7 accidents is

$$P(A) = P(A|T)P(T) + P(A|\bar{T})P(\bar{T}) = \frac{e^{-10}10^7}{7!} \cdot 0.6 + \frac{e^{-4}4^7}{7!} \cdot 0.4.$$

Therefore, the conditional probability that there was a thunderstorm given that there were 7 accidents yesterday is

$$P(T|A) = \frac{\frac{e^{-10}10^7}{7!} \cdot 0.6}{\frac{e^{-10}10^7}{7!} \cdot 0.6 + \frac{e^{-4}4^7}{7!} \cdot 0.4}.$$

3.37 An interactive system consists of ten terminals that are connected to the central computer. At any time, each terminal is ready to transmit a message with probability 0.7, independently of other terminals. Find the probability that exactly 6 terminals are ready to transmit at 8 o'clock.

Solution. Let X be the number of terminals that are ready to transmit at 8 o'clock. Then X follows a binomial distribution with parameters $n = 10$ and $p = 0.7$. The probability that exactly 6 terminals are ready to transmit is

$$P(X = 6) = \binom{10}{6}(0.7)^6(0.3)^4.$$

3.38 Network breakdowns are unexpected rare events that occur every 3 weeks, on the average. Compute the probability of more than 4 breakdowns during a 21-week period.

Solution. Let X be the number of network breakdowns during a 21-week period. Then X follows a Poisson distribution with parameter $\lambda = \frac{21}{3} = 7$. The probability of more than 4 breakdowns is

$$P(X > 4) = 1 - P(X \leq 4) = 1 - \sum_{k=0}^{4} P(X = k) = 1 - \sum_{k=0}^{4} \frac{e^{-7}7^k}{k!}.$$

We can use Table A3 of the textbook to find that $P(X \leq 4) = F(4) \approx 0.173$. Therefore, the probability of more than 4 breakdowns during a 21-week period is approximately $1 - 0.173 = 0.827$.

Chapter 4

4.1 The lifetime, in years, of some electronic component is a continuous random variable with the density

$$f(x) = \begin{cases} \frac{k}{x^4}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

Find k, the cumulative distribution function, and the probability for the lifetime to exceed 2 years.

Solution. Since $f(x)$ is a probability density function, we have

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Calculating the integral, we get

$$\int_1^{\infty} \frac{k}{x^4}\,dx = k\int_1^{\infty} x^{-4}\,dx = k\left[\frac{x^{-3}}{-3}\right]_1^{\infty} = k\left(0-\left(-\frac{1}{3}\right)\right) = \frac{k}{3}.$$

Setting this equal to 1, we find $k = 3$. The cumulative distribution function is

$$F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_1^x \frac{3}{t^4}\,dt = 3\left[\frac{t^{-3}}{-3}\right]_1^x = 1 - \frac{1}{x^3}, \quad x \geq 1.$$

The probability for the lifetime to exceed 2 years is

$$P(X > 2) = 1 - F(2) = 1 - \left(1 - \frac{1}{2^3}\right) = \frac{1}{8}.$$

4.4 Lifetime of a certain hardware is a continuous random variable with density

$$f(x) = \begin{cases} K - \frac{x}{50}, & \text{for } 0 < x < 10 \text{ years} \\ 0, & \text{for all other } x \end{cases}$$

(a) Find K.

(b) What is the probability of a failure within the first 5 years?

(c) What is the expectation of the lifetime?

Solution.

(a) Since $f(x)$ is a probability density function, we have

$$\int_0^{10} \left(K - \frac{x}{50}\right)\,dx = 1.$$

Calculating the integral, we get

$$Kx - \frac{x^2}{100}\bigg|_0^{10} = 10K - 1 = 1.$$

Setting this equal to 1, we find $10K - 1 = 1 \implies 10K = 2 \implies K = 0.2$.

(b) The probability of a failure within the first 5 years is

$$P(X < 5) = \int_0^5 \left(0.2 - \frac{x}{50}\right)\,dx = 0.2x - \frac{x^2}{100}\bigg|_0^5 = 1 - 0.25 = 0.75.$$

(c) The expectation of the lifetime is

$$\mathbb{E}(X) = \int_0^{10} x\left(0.2 - \frac{x}{50}\right)\,dx = \int_0^{10} \left(0.2x - \frac{x^2}{50}\right)\,dx = 0.1x^2 - \frac{x^3}{150}\bigg|_0^{10} = \frac{10}{3} \text{ years.}$$

4.7 The time it takes a printer to print a job is an Exponential random variable with the expectation of 12 seconds. You send a job to the printer at 10:00 am, and it appears to be third in line. What is the probability that your job will be ready before 10:01?

Solution. Let T be the time it takes for the printer to print a job. Since T is an Exponential random variable with expectation 12 seconds, the rate parameter λ is given by

$$\lambda = \frac{1}{\mathbb{E}(T)} = \frac{1}{12} \text{ seconds}^{-1}.$$

Let X be the number of jobs completed in 60 seconds. Then X follows a Poisson distribution with parameter $\lambda t = \frac{1}{12} \cdot 60 = 5$. The probability that your job will be ready before 10:01 is the probability that at least 3 jobs are completed in 60 seconds, which is

$$P(X \geq 3) = 1 - P(X < 3) = 1 - \sum_{k=0}^{2} P(X = k) = 1 - \sum_{k=0}^{2} \frac{e^{-5}5^k}{k!}.$$

Let F be cumulative distribution function of the Poisson distribution with parameter $\lambda = 5$. From Table A3, we find that $F(2) \approx 0.125$. Therefore, the probability that your job will be ready before 10:01 is approximately $1 - 0.125 = 0.875$.

4.9 On the average, a computer experiences breakdowns every 5 months. The time until the first breakdown and the times between any two consecutive breakdowns are independent Exponential random variables. After the third breakdown, a computer requires a special maintenance.

(a) Compute the probability that a special maintenance is required within the next 9 months.

(b) Given that a special maintenance was not required during the first 12 months, what is the probability that it will not be required within the next 4 months?

Solution.

(a) Let X be the number of breakdowns in 9 months. Then X follows a Poisson distribution with parameter $\lambda = \frac{9}{5} = 1.8$. The probability that a special maintenance is required within the next 9 months is the probability that at least 3 breakdowns occur in 9 months, which is

$$P(X \geq 3) = 1 - P(X < 3) = 1 - \sum_{k=0}^{2} P(X = k) = 1 - \sum_{k=0}^{2} \frac{e^{-1.8}1.8^k}{k!}.$$

Let F be cumulative distribution function of the Poisson distribution with parameter $\lambda = 1.8$. From Table A3, we find that $F(2) \approx 0.731$. Therefore, the probability that a special maintenance is required within the next 9 months is approximately $1 - F(2) = 1 - 0.731 = 0.269$.

(b) Given that a special maintenance was not required during the first 12 months, we want to find the probability that it will not be required within the next 4 months. Let T be the time until the third breakdown. Then T follows a Gamma distribution with shape parameter $k = 3$ and rate parameter $\lambda = \frac{1}{5}$. We have by Bayes' rule that

$$P(T > 16 | T > 12) = \frac{P(T > 16, T > 12)}{P(T > 12)} = \frac{P(T > 16)}{P(T > 12)}.$$

To find $P(T > 12)$, we consider $X \sim \text{Poisson}(\lambda = \frac{12}{5} = 2.4)$. Then

$$P(T > 12) = P(X < 3) = F_X(2) = 0.570,$$

where F_X is the cumulative distribution function of the Poisson distribution with parameter $\lambda = 2.4$. Similarly, to find $P(T > 16)$, we consider $Y \sim \text{Poisson}(\lambda = \frac{16}{5} = 3.2)$. Then

$$P(T > 16) = P(Y < 3) = P(Y \leq 2) = \sum_{y=0}^{2} \frac{e^{-3.2} 3.2^y}{y!} = 0.423,$$

which is directly computed since Table A3 does not provide the value for $\lambda = 3.2$. Therefore, we have

$$P(T > 16 | T > 12) = \frac{P(T > 16)}{P(T > 12)} = \frac{0.423}{0.570} \approx 0.742.$$

4.10 Two computer specialists are completing work orders. The first specialist receives 60% of all orders. Each order takes her Exponential amount of time with parameter $\lambda_1 = 3hrs^{-1}$. The second specialist receives the remaining 40% of orders. Each order takes him Exponential amount of time with parameter $\lambda_2 = 2hrs^{-1}$. A certain order was submitted 30 minutes ago, and it is still not ready. What is the probability that the first specialist is working on it?

Solution. Let T_1 be the time it takes for the first specialist to complete an order, and let T_2 be the time it takes for the second specialist to complete an order. We want to find $P(S_1 | T > 0.5)$, where S_1 is the event that the first specialist is working on the order, and T is the time until the order is completed. By Bayes' rule, we have

$$P(S_1 | T > 0.5) = \frac{P(T > 0.5 | S_1) P(S_1)}{P(T > 0.5)}.$$

We know that $P(S_1) = 0.6$ and $P(S_2) = 0.4$, where S_2 is the event that the second specialist is working on the order. The probability that the order takes more than 0.5 hours given that the first specialist is working on it is

$$P(T > 0.5 | S_1) = e^{-\lambda_1 \cdot 0.5} = e^{-3 \cdot 0.5} = e^{-1.5} \approx 0.2231,$$

and the probability that the order takes more than 0.5 hours given that the second specialist is working on it is

$$P(T > 0.5 | S_2) = e^{-\lambda_2 \cdot 0.5} = e^{-2 \cdot 0.5} = e^{-1} \approx 0.3679.$$

The total probability that the order takes more than 0.5 hours is

$$\begin{aligned}
P(T > 0.5) &= P(T > 0.5 | S_1) P(S_1) + P(T > 0.5 | S_2) P(S_2) \\
&= 0.2231 \cdot 0.6 + 0.3679 \cdot 0.4 \\
&\approx 0.13386 + 0.14716 \\
&\approx 0.28102.
\end{aligned}$$

Therefore, the conditional probability that the first specialist is working on the order given that it is still not ready after 30 minutes is

$$P(S_1 | T > 0.5) = \frac{0.2231 \cdot 0.6}{0.28102} \approx \frac{0.13386}{0.28102} \approx 0.476.$$

4.11 Consider a satellite whose work is based on a certain block A. This block has an independent backup B. The satellite performs its task until both A and B fail. The lifetimes of A and B are exponentially distributed with the mean lifetime of 10 years.

(a) What is the probability that the satellite will work for more than 10 years?

(b) Compute the expected lifetime of the satellite.

Solution.

(a) Let T_A and T_B be the lifetimes of blocks A and B, respectively. Since both lifetimes are independent and exponentially distributed with mean 10 years, the rate parameter λ is given by

$$\lambda = \frac{1}{10} \text{ years}^{-1}.$$

The satellite will work for more than 10 years if at least one of the blocks is still functioning after 10 years. The probability that both blocks fail within 10 years is

$$\begin{aligned} P(T_A \leq 10 \text{ and } T_B \leq 10) &= P(T_A \leq 10)P(T_B \leq 10) = (1 - e^{-\lambda \cdot 10})^2 \\ &= (1 - e^{-1})^2 \approx (1 - 0.3679)^2 \approx 0.3996. \end{aligned}$$

Therefore, the probability that the satellite will work for more than 10 years is

$$P(T_A > 10 \text{ or } T_B > 10) = 1 - P(T_A \leq 10 \text{ and } T_B \leq 10) \approx 1 - 0.3996 \approx 0.6004.$$

(b) Let $T = \max\{T_A, T_B\}$ be the lifetime of the satellite. Then the cumulative distribution function of T is given by

$$F_T(t) = P(T \leq t) = P(T_A \leq t \text{ and } T_B \leq t) = P(T_A \leq t)P(T_B \leq t) = (1 - e^{-\lambda t})^2.$$

The probability density function of T is then given by

$$f_T(t) = \frac{d}{dt}F_T(t) = 2(1 - e^{-\lambda t})e^{-\lambda t}\lambda = 2\lambda e^{-\lambda t}(1 - e^{-\lambda t}).$$

Since $\lambda = \frac{1}{10}$, we have

$$f_T(t) = \frac{1}{5}e^{-\frac{t}{10}}(1 - e^{-\frac{t}{10}}).$$

The expected lifetime of the satellite is the expected value of T, which can be computed as

$$\begin{aligned} \mathbb{E}[T] &= \int_0^\infty t f_T(t)\, dt = \int_0^\infty t \cdot 2\lambda e^{-\lambda t}(1 - e^{-\lambda t})\, dt = 2\lambda \int_0^\infty te^{-\lambda t}(1 - e^{-\lambda t})\, dt \\ &= 2\lambda \int_0^\infty te^{-\lambda t}\, dt - 2\lambda \int_0^\infty te^{-2\lambda t}\, dt. \end{aligned}$$

Calculating these integrals, we get

$$\int_0^\infty te^{-\lambda t}\, dt = \frac{1}{\lambda^2}, \quad \int_0^\infty te^{-2\lambda t}\, dt = \frac{1}{(2\lambda)^2} = \frac{1}{4\lambda^2}.$$

Therefore, we have

$$\mathbb{E}[T] = 2\lambda \left(\frac{1}{\lambda^2} - \frac{1}{4\lambda^2}\right) = 2\lambda \left(\frac{3}{4\lambda^2}\right) = \frac{3}{2\lambda} = \frac{3}{2} \cdot 10 = 15 \text{ years}.$$

This is another way to solve this problem. The expected lifetime of the satellite is the expected value of the maximum of the two independent lifetimes, which can be computed as

$$\begin{aligned}\mathbb{E}[\max(T_A, T_B)] &= \int_0^\infty P(\max(T_A, T_B) > t)\,dt = \int_0^\infty (1 - P(T_A \le t)P(T_B \le t))\,dt \\ &= \int_0^\infty (1 - (1 - e^{-\lambda t})^2)\,dt.\end{aligned}$$

Calculating this integral, we get

$$\begin{aligned}\mathbb{E}[\max(T_A, T_B)] &= \int_0^\infty (2e^{-\lambda t} - e^{-2\lambda t})\,dt = 2\int_0^\infty e^{-\lambda t}\,dt - \int_0^\infty e^{-2\lambda t}\,dt \\ &= 2 \cdot \frac{1}{\lambda} - \frac{1}{2\lambda} = \frac{4}{2\lambda} - \frac{1}{2\lambda} = \frac{3}{2\lambda} = \frac{3}{2} \cdot 10 = 15 \text{ years.}\end{aligned}$$

4.16 Let Z be a Standard Normal random variable. Compute (a) $P(Z < 1.25)$ (b) $P(Z \le 1.25)$ (c) $P(Z > 1.25)$ (d) $P(|Z| \le 1.25)$ (e) $P(Z < 6.0)$ (f) $P(Z > 6.0)$ (g) With probability 0.8, variable Z does not exceed what value?

Solution.

(a) $P(Z < 1.25) = 0.8944$ (from Standard Normal table)

(b) $P(Z \le 1.25) = 0.8944$ (from Standard Normal table)

(c) $P(Z > 1.25) = 1 - P(Z \le 1.25) = 1 - 0.8944 = 0.1056$

(d) $P(|Z| \le 1.25) = P(-1.25 \le Z \le 1.25) = P(Z \le 1.25) - P(Z < -1.25) = 0.8944 - (1 - 0.8944) = 0.8944 - 0.1056 = 0.7888$

(e) $P(Z < 6.0) = 1$ (since 6.0 is far in the tail of the Standard Normal distribution)

(f) $P(Z > 6.0) = 0$ (since 6.0 is far in the tail of the Standard Normal distribution)

(g) To find the value z such that $P(Z \le z) = 0.8$, we look up the Standard Normal table and find that $z \approx 0.8416$.

4.17 For a Standard Normal random variable Z, compute (a) $P(Z \ge 0.99)$ (b) $P(Z < -0.99)$ (c) $P(Z \le 0.99)$ (d) $P(|Z| > 0.99)$ (e) $P(Z < 10.0)$ (f) $P(Z > 10.0)$ (g) With probability 0.9, variable Z is less than what?

Solution.

(a) $P(Z \ge 0.99) = 1 - P(Z \le 0.99) = 1 - 0.8389 = 0.1611$ (from Standard Normal table)

(b) $P(Z < -0.99) = 1 - P(Z \le 0.99) = 1 - 0.8389 = 0.1611$ (from Standard Normal table)

(c) $P(Z \le 0.99) = 0.8389$ (from Standard Normal table)

(d) $P(|Z| > 0.99) = P(Z < -0.99) + P(Z > 0.99) = (1 - P(Z \le 0.99)) + (1 - P(Z \le 0.99)) = 2(1 - 0.8389) = 2(0.1611) = 0.3222$

(e) $P(Z < 10.0) = 1$ (since 10.0 is far in the tail of the Standard Normal distribution)

(f) $P(Z > 10.0) = 0$ (since 10.0 is far in the tail of the Standard Normal distribution)

(g) To find the value z such that $P(Z \le z) = 0.9$, we look up the Standard Normal table and find that $z \approx 1.2816$.

4.18 For a Normal random variable X with $\mathbb{E}(X) = -3$ and $\mathrm{Var}(X) = 4$, compute

(a) $P(X \le 2.39)$

(b) $P(Z \ge -2.39)$

(c) $P(|X| \ge 2.39)$

(d) $P(|X+3| \ge 2.39)$

(e) $P(X < 5)$

(f) $P(|X| < 5)$

(g) With probability 0.33, variable X exceeds what value?

Solution. Let Z be a Standard Normal random variable. We can standardize X as

$$Z = \frac{X-\mu}{\sigma} = \frac{X+3}{2}.$$

(a) $P(X \le 2.39) = P\left(Z \le \frac{2.39+3}{2}\right) = P(Z \le 2.695) \approx 0.9965$ (from Standard Normal table).

(b) $P(Z \ge -2.39) = P(Z \ge -2.39) = 1 - P(Z < -2.39) = 1 - (1 - P(Z \le 2.39)) = P(Z \le 2.39) \approx 0.9916$ (from Standard Normal table).

(c) $P(|X| \ge 2.39) = P(X \le -2.39) + P(X \ge 2.39) = P\left(Z \le \frac{-2.39+3}{2}\right) + P\left(Z \ge \frac{2.39+3}{2}\right) = P(Z \le 0.305) + P(Z \ge 2.695) \approx 0.6207 + (1 - 0.9965) = 0.6207 + 0.0035 = 0.6242$.

(d) $P(|X+3| \ge 2.39) = P(X+3 \le -2.39) + P(X+3 \ge 2.39) = P(X \le -5.39) + P(X \ge -0.61) = P\left(Z \le \frac{-5.39+3}{2}\right) + P\left(Z \ge \frac{-0.61+3}{2}\right) = P(Z \le -1.195) + P(Z \ge 1.195) \approx (1 - 0.8849) + (1 - 0.8849) = 0.1151 + 0.1151 = 0.2302$ (from Standard Normal table).

(e) $P(X < 5) = P\left(Z < \frac{5+3}{2}\right) = P(Z < 4) \approx 1$ (from Standard Normal table).

(f) $P(|X| < 5) = P(-5 < X < 5) = P\left(\frac{-5+3}{2} < Z < \frac{5+3}{2}\right) = P(-1 < Z < 4) = P(Z < 4) - P(Z < -1) \approx 1 - (1 - 0.8413) = 0.8413$ (from Standard Normal table).

(g) To find the value x such that $P(X > x) = 0.33$, we first find the corresponding value z such that $P(Z > z) = 0.33$. From the Standard Normal table, we find that $z \approx 0.44$. Then,

$$x = \mu + z\sigma = -3 + (0.44)(2) = -3 + 0.88 = -2.12.$$

Therefore, with probability 0.33, variable X exceeds the value -2.12.

4.19 According to one of the Western Electric rules for quality control, a produced item is considered conforming if its measurement falls within three standard deviations from the target value. Suppose that the process is in control so that the expected value of each measurement equals the target value. What percent of items will be considered conforming, if the distribution of measurements is

(a) $\mathrm{Normal}(\mu, \sigma)$?

(b) $\mathrm{Uniform}(a, b)$?

Solution.

(a) For a Normal(μ, σ) distribution, the probability that a measurement falls within three standard deviations from the mean is

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) = P(-3 \le Z \le 3) = P(Z \le 3) - P(Z < -3)$$
$$\approx 0.99865 - (1 - 0.99865) = 0.9973.$$

(b) For a Uniform(a, b) distribution, $\mu = \frac{a+b}{2}$ and $\sigma = \frac{b-a}{2\sqrt{3}}$. The probability that a measurement falls within three standard deviations from the mean is

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) = P\left(\frac{a+b}{2} - 3 \cdot \frac{b-a}{2\sqrt{3}} \le X \le \frac{a+b}{2} + 3 \cdot \frac{b-a}{2\sqrt{3}}\right)$$
$$= P\left(\frac{a+b}{2} - \frac{3(b-a)}{2\sqrt{3}} \le X \le \frac{a+b}{2} + \frac{3(b-a)}{2\sqrt{3}}\right).$$

Since the range of the Uniform distribution is $[a, b]$, we need to check if the interval $\left[\frac{a+b}{2} - \frac{3(b-a)}{2\sqrt{3}}, \frac{a+b}{2} + \frac{3(b-a)}{2\sqrt{3}}\right]$ is within $[a, b]$. If it is, then the probability is 1. If not, we need to calculate the proportion of the interval that falls within $[a, b]$. The length of the interval is $\frac{3(b-a)}{\sqrt{3}} = \sqrt{3}(b-a)$. The length of the Uniform distribution is $b - a$, which is full covered. Therefore the probabilty $P(\mu - 3\sigma \le X \le \mu + 3\sigma) = 1$.

4.20 Refer to Exercise 4.19. What percent of items falls beyond 1.5 standard deviations from the mean, if the distribution of measurements is

(a) Normal(μ, σ)?

(b) Uniform(a, b)?

Solution.

(a) For a Normal(μ, σ) distribution, the probability that a measurement falls beyond 1.5 standard deviations from the mean is (we normalize X to $Z = \frac{X-\mu}{\sigma}$ which follows a Standard Normal distribution):

$$P(|X - \mu| > 1.5\sigma) = P(X < \mu - 1.5\sigma) + P(X > \mu + 1.5\sigma)$$
$$= P(Z < -1.5) + P(Z > 1.5)$$
$$\approx 0.0668 + 0.0668 = 0.1336.$$

(b) For a Uniform(a, b) distribution, $\mu = \frac{a+b}{2}$ and $\sigma = \frac{b-a}{2\sqrt{3}}$. The probability that a measurement falls within three standard deviations from the mean is

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) = P\left(\frac{a+b}{2} - 1.5 \cdot \frac{b-a}{2\sqrt{3}} \le X \le \frac{a+b}{2} + 1.5 \cdot \frac{b-a}{2\sqrt{3}}\right)$$
$$= P\left(\frac{a+b}{2} - \frac{1.5(b-a)}{2\sqrt{3}} \le X \le \frac{a+b}{2} + \frac{1.5(b-a)}{2\sqrt{3}}\right).$$

Since the range of the Uniform distribution is $[a, b]$, we need to check if the interval $\left[\frac{a+b}{2} - \frac{1.5(b-a)}{2\sqrt{3}}, \frac{a+b}{2} + \frac{1.5(b-a)}{2\sqrt{3}}\right]$ is within $[a, b]$. If it is, then the probability is 1. If not, we need to calculate the proportion of the interval that falls within $[a, b]$. The length of the interval is $\frac{1.5(b-a)}{\sqrt{3}} = \frac{\sqrt{3}}{2}(b-a)$ which is included in $b - a$. Therefore the probabilty $P(|X - \mu| > 1.5\sigma) = 1 - \frac{\sqrt{3}}{2} = 1 - 0.866 = 0.134$.

4.21 The average height of professional basketball players is around 6 feet 7 inches, and the standard deviation is 3.89 inches. Assuming Normal distribution of heights within this group,

(a) What percent of professional basketball players are taller than 7 feet?

(b) If your favorite player is within the tallest 20% of all players, what can his height be?

Solution.

(a) Let X be the height of professional basketball players. We want to find $P(X > 7 \text{ feet})$. First, we convert 7 feet to inches: 7 feet $=$ 84 inches. The mean height is $\mu = 6$ feet 7 inches $= 79$ inches, and the standard deviation is $\sigma = 3.89$ inches. We standardize the height:

$$Z = \frac{X - \mu}{\sigma} = \frac{84 - 79}{3.89} \approx 1.285.$$

Now, we find $P(Z > 1.285) = 1 - P(Z \leq 1.285) \approx 1 - 0.8997 = 0.1003$. Therefore, approximately 10.03% of professional basketball players are taller than 7 feet.

(b) We want to find the height corresponding to the tallest 20% of players. This means we need to find the 80th percentile of the Normal distribution. From the Standard Normal table, we find that $P(Z \leq z) = 0.8$ corresponds to $z \approx 0.8416$. Now, we convert this back to the original height:

$$X = \mu + z\sigma = 79 + 0.8416 \cdot 3.89 \approx 79 + 3.27 \approx 82.27 \text{ inches}$$

4.22 Refer to the country in Example 4.11 on p. 91, where household incomes follow Normal distribution with $\mu = 900$ coins and $\sigma = 200$ coins.

(a) A recent economic reform made households with the income below 640 coins qualify for a free bottle of milk at every breakfast. What portion of the population qualifies for a free bottle of milk?

(b) Moreover, households with an income within the lowest 5% of the population are entitled to a free sandwich. What income qualifies a household to receive free sandwiches?

Solution.

(a) We want to find $P(X < 640)$, where X is the household income. We standardize:

$$Z = \frac{X - \mu}{\sigma} = \frac{640 - 900}{200} = -1.3.$$

Now, we find $P(Z < -1.3) \approx 0.0968$. Therefore, approximately 9.68% of the population qualifies for a free bottle of milk.

(b) We want to find the income corresponding to the lowest 5% of the population. This means we need to find the 5th percentile of the Normal distribution. From the Standard Normal table, we find that $P(Z \leq z) = 0.05$ corresponds to $z \approx -1.645$. Now, we convert this back to the original income:

$$X = \mu + z\sigma = 900 + (-1.645) \cdot 200 \approx 900 - 329 = 571.$$

So households with an income below 571 coins qualify for a free sandwich.

4.23 The lifetime of a certain electronic component is a random variable with the expectation of 5000 hours and a standard deviation of 100 hours. What is the probability that the average lifetime of 400 components is less than 5012 hours?

Solution. Let X be the lifetime of a single electronic component. We have $\mathbb{E}(X) = 5000$ hours and $\sigma_X = 100$ hours. The average lifetime of 400 components is given by

$$\bar{X} = \frac{1}{400} \sum_{i=1}^{400} X_i.$$

By the Central Limit Theorem, $\bar{X}$ is approximately normally distributed with mean $\mu_{\bar{X}} = \mu_X = 5000$ hours and standard deviation $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{100}{\sqrt{400}} = 5$ hours. We want to find $P(\bar{X} < 5012)$. We standardize:

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{5012 - 5000}{5} = \frac{12}{5} = 2.4.$$

Now, we find $P(Z < 2.4) \approx 0.9918$. Therefore, the probability that the average lifetime of 400 components is less than 5012 hours is approximately 0.9918.

4.24 Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download one file, with a variance of 16 sec^2. What is the probability that the software is installed in less than 20 minutes?

Solution. Let X_i be the time to download the i-th file, where $i = 1, 2, \ldots, 82$. We have $\mathbb{E}(X_i) = 15$ sec and $\sigma^2_{X_i} = 16$ sec^2. The total time to download all 82 files is given by

$$T = \sum_{i=1}^{82} X_i.$$

By the Central Limit Theorem, T is approximately normally distributed with mean $\mu_T = n \cdot \mu_{X_i} = 82 \cdot 15 = 1230$ sec and standard deviation $\sigma_T = \sqrt{n} \cdot \sigma_{X_i} = \sqrt{82} \cdot 4 \approx 36.22$ sec. We want to find $P(T < 20 \text{ minutes}) = P(T < 1200 \text{ sec})$. We standardize:

$$Z = \frac{T - \mu_T}{\sigma_T} = \frac{1200 - 1230}{36.22} \approx -0.828.$$

Now, we find $P(Z < -0.828) \approx 0.204$. Therefore, the probability that the software is installed in less than 20 minutes is approximately 0.204.

4.25 Among all the computer chips produced by a certain factory, 6 percent are defective. A sample of 400 chips is selected for inspection.

(a) What is the probability that this sample contains between 20 and 25 defective chips (including 20 and 25)?

(b) Suppose that each of 40 inspectors collects a sample of 400 chips. What is the probability that at least 8 inspectors will find between 20 and 25 defective chips in their samples?

Solution.

(a) Let X be the number of defective chips in the sample of 400 chips. We have $X \sim \text{Binomial}(n = 400, p = 0.06)$. The mean and variance of X are given by

$$\mu_X = np = 400 \cdot 0.06 = 24,$$

$$\sigma_X^2 = np(1-p) = 400 \cdot 0.06 \cdot 0.94 = 22.56.$$

Since n is large, we can approximate the binomial distribution by a normal distribution with mean $\mu_X = 24$ and variance $\sigma_X^2 = 22.56$. We want to find $P(20 \leq X \leq 25)$. With a continuity correction, we find $P(19.5 \leq X \leq 25.5)$ instead and get:

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{19.5 - 24}{\sqrt{22.56}} \approx -0.95,$$

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{25.5 - 24}{\sqrt{22.56}} \approx 0.32.$$

Therefore, $P(19.5 \leq X \leq 25.5) = P(-0.95 \leq Z \leq 0.32) = P(Z \leq 0.32) - P(Z \leq -0.95) \approx 0.6255 - 0.1711 = 0.4544$. Thus, the probability that the sample contains between 20 and 25 defective chips is approximately 0.4544.

(b) Let Y be the number of inspectors who find between 20 and 25 defective chips in their samples. We have $Y \sim \text{Binomial}(n = 40, p = 0.4544)$. The mean and variance of Y are given by

$$\mu_Y = np = 40 \cdot 0.4544 \approx 18.176,$$

$$\sigma_Y^2 = np(1-p) = 40 \cdot 0.4544 \cdot (1 - 0.4544) \approx 9.$$

Since n is large, we can approximate the binomial distribution by a normal distribution with mean $\mu_Y \approx 18.176$ and variance $\sigma_Y^2 \approx 9$. We want to find $P(Y \geq 8)$. With a continuity correction, we find $P(Y \geq 7.5)$ instead and get:

$$Z = \frac{Y - \mu_Y}{\sigma_Y} = \frac{7.5 - 18.176}{\sqrt{9}} \approx -3.56.$$

Therefore, $P(Y \geq 7.5) = P(Z \geq -3.56) = 1 - P(Z < -3.56) \approx 1 - 0.0002 = 0.9998$. Thus, the probability that at least 8 inspectors will find between 20 and 25 defective chips in their samples is approximately 0.9998.

4.26 An average scanned image occupies 0.6 megabytes of memory with a standard deviation of 0.4 megabytes. If you plan to publish 80 images on your web site, what is the probability that their total size is between 47 megabytes and 50 megabytes?

Solution. Let X_i be the size of the i-th scanned image, where $i = 1, 2, \ldots, 80$. We have $\mathbb{E}(X_i) = 0.6$ megabytes and $\sigma_{X_i} = 0.4$ megabytes. The total size of the 80 images is given by

$$S = \sum_{i=1}^{80} X_i.$$

By the Central Limit Theorem, S is approximately normally distributed with mean $\mu_S = n \cdot \mu_{X_i} = 80 \cdot 0.6 = 48$ megabytes and standard deviation $\sigma_S = \sqrt{n} \cdot \sigma_{X_i} = \sqrt{80} \cdot 0.4 \approx 3.5777$ megabytes. We want to find $P(47 \leq S \leq 50)$. Standardizing, we get

$$Z_1 = \frac{47 - 48}{3.5777} \approx -0.2795,$$

$$Z_2 = \frac{50 - 48}{3.5777} \approx 0.5590.$$

Therefore, $P(47 \leq S \leq 50) = P(-0.2795 \leq Z \leq 0.5590) = P(Z \leq 0.5590) - P(Z \leq -0.2795) \approx 0.7123 - 0.3890 = 0.3233$. Thus, the probability that the total size of the 80 images is between 47 megabytes and 50 megabytes is approximately 0.3233.

4.27 A certain computer virus can damage any file with probability 35%, independently of other files. Suppose this virus enters a folder containing 2400 files. Compute the probability that between 800 and 850 files get damaged (including 800 and 850).

Solution. Let X be the number of files that get damaged. We have $X \sim \text{Binomial}(n = 2400, p = 0.35)$. The mean and variance of X are given by

$$\mu_X = np = 2400 \cdot 0.35 = 840,$$

$$\sigma_X^2 = np(1-p) = 2400 \cdot 0.35 \cdot 0.65 = 546.$$

Since n is large, we can approximate the binomial distribution by a normal distribution with mean $\mu_X = 840$ and variance $\sigma_X^2 = 546$. We want to find $P(800 \leq X \leq 850)$. With a continuity correction, we find $P(799.5 \leq X \leq 850.5)$ instead and get

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{799.5 - 840}{\sqrt{546}} \approx -1.73.$$

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{850.5 - 840}{\sqrt{546}} \approx 0.44.$$

Therefore, $P(799.5 \leq X \leq 850.5) = P(-1.73 \leq Z \leq 0.44) = P(Z \leq 0.44) - P(Z \leq -1.73) \approx 0.6700 - 0.0418 = 0.6282$. Thus, the probability that between 800 and 850 files get damaged is approximately 0.6282.

4.30 Seventy independent messages are sent from an electronic transmission center. Messages are processed sequentially, one after another. Transmission time of each message is Exponential with parameter $\lambda = 5\text{min}^{-1}$. Find the probability that all 70 messages are transmitted in less than 12 minutes. Use the Central Limit Theorem.

Solution. Let T_i be the transmission time of the i-th message, where $i = 1, 2, \ldots, 70$. We have $T_i \sim \text{Exponential}(\lambda = 5)$, so the mean and variance of T_i are given by

$$\mu_{T_i} = \frac{1}{\lambda} = \frac{1}{5} = 0.2 \text{ minutes},$$

$$\sigma_{T_i}^2 = \frac{1}{\lambda^2} = \frac{1}{25} = 0.04 \text{ minutes}^2.$$

The total transmission time for all 70 messages is given by

$$S = \sum_{i=1}^{70} T_i.$$

By the Central Limit Theorem, S is approximately normally distributed with mean $\mu_S = n \cdot \mu_{T_i} = 70 \cdot 0.2 = 14$ minutes and standard deviation $\sigma_S = \sqrt{n} \cdot \sigma_{T_i} = \sqrt{70} \cdot 0.2 \approx 1.6733$ minutes. We want to find $P(S < 12)$. Standardizing, we get

$$Z = \frac{12 - 14}{1.6733} \approx -1.195.$$

Therefore, $P(S < 12) = P(Z < -1.195) \approx 0.1165$. Thus, the probability that all 70 messages are transmitted in less than 12 minutes is approximately 0.1165.

4.31 A computer lab has two printers. Printer I handles 40% of all the jobs. Its printing time is Exponential with the mean of 2 minutes. Printer II handles the remaining 60% of jobs. Its printing time is Uniform between 0 minutes and 5 minutes. A job was printed in less than 1 minute. What is the probability that it was printed by Printer I?

Solution. Let P_1 be the event that the job was printed by Printer I, and P_2 be the event that the job was printed by Printer II. We are given that $P(P_1) = 0.4$ and $P(P_2) = 0.6$. Let T be the printing time of the job. We want to find $P(P_1|T < 1)$. By Bayes' rule, we have

$$P(P_1|T < 1) = \frac{P(T < 1|P_1)P(P_1)}{P(T < 1)}.$$

We need to compute $P(T < 1|P_1)$ and $P(T < 1|P_2)$. For Printer I, we have

$$P(T < 1|P_1) = 1 - e^{-\lambda t} = 1 - e^{-t/2} \text{ for } t = 1 \Rightarrow P(T < 1|P_1) = 1 - e^{-1/2} \approx 0.3935.$$

For Printer II, we have

$$P(T < 1|P_2) = \frac{1-0}{5-0} = \frac{1}{5} = 0.2.$$

Now we can compute $P(T < 1)$:

$$P(T < 1) = P(T < 1|P_1)P(P_1) + P(T < 1|P_2)P(P_2) = 0.3935 \cdot 0.4 + 0.2 \cdot 0.6 = 0.2774.$$

Therefore,

$$P(P_1|T < 1) = \frac{P(T < 1|P_1)P(P_1)}{P(T < 1)} = \frac{0.3935 \cdot 0.4}{0.2774} \approx 0.568.$$

Thus, the probability that the job was printed by Printer I, given that it was printed in less than 1 minute, is approximately 0.568.

4.33 Upgrading a certain software package requires installation of 68 new files. Files are installed consecutively. The installation time is random, but on the average, it takes 15 sec to install one file, with a variance of 11 sec^2.

(a) What is the probability that the whole package is upgraded in less than 12 minutes?

(b) A new version of the package is released. It requires only N new files to be installed, and it is promised that 95% of the time upgrading takes less than 10 minutes. Given this information, compute N.

Solution.

(a) Let T_i be the installation time of the i-th file, where $i = 1, 2, \ldots, 68$. We have $\mathbb{E}(T_i) = 15$ sec and $\sigma^2_{T_i} = 11$ sec^2. The total installation time for all 68 files is given by

$$S = \sum_{i=1}^{68} T_i.$$

By the Central Limit Theorem, S is approximately normally distributed with mean $\mu_S = n \cdot \mu_{T_i} = 68 \cdot 15 = 1020$ sec and standard deviation $\sigma_S = \sqrt{n} \cdot \sigma_{T_i} = \sqrt{68} \cdot \sqrt{11} \approx 27.29$ sec. We want to find $P(S < 12 \text{ minutes}) = P(S < 720 \text{ sec})$. Standardizing, we get

$$Z = \frac{720 - 1020}{27.29} \approx -10.99.$$

Therefore, $P(S < 720) = P(Z < -10.99) \approx 0$. Thus, the probability that the whole package is upgraded in less than 12 minutes is approximately 0.

(b) We want to find N such that $P(S < 10 \text{ minutes}) = 0.95$. Let S_N be the total installation time for N files. We have $\mu_{S_N} = N \cdot 15$ sec and $\sigma_{S_N} = \sqrt{N} \cdot \sqrt{11}$ sec. We want to find N such that

$$P(S_N < 600 \text{ sec}) = 0.95.$$

Standardizing, we get

$$Z = \frac{600 - N \cdot 15}{\sqrt{N} \cdot \sqrt{11}}.$$

We want $P(Z < z) = 0.95$, which corresponds to $z \approx 1.645$. Therefore, we have

$$\frac{600 - N \cdot 15}{\sqrt{N} \cdot \sqrt{11}} = 1.645.$$

Rearranging, we get

$$600 - 15N = 1.645 \cdot \sqrt{11N}.$$

Solving this quadratic equation of $\sqrt{N}$, we get $N \approx 38$. Therefore, the new version of the package requires approximately 38 new files to be installed to ensure that 95% of the time upgrading takes less than 10 minutes.

Chapter 8

8.1 The numbers of blocked intrusion attempts on each day during the first two weeks of the month were

56, 47, 49, 37, 38, 60, 50, 43, 43, 59, 50, 56, 54, 58

After the change of firewall settings, the numbers of blocked intrusions during the next 20 days were

53, 21, 32, 49, 45, 38, 44, 33, 32, 43, 53, 46, 36, 48, 39, 35, 37, 36, 39, 45.

Comparing the number of blocked intrusions before and after the change,

(a) construct side-by-side stem-and-leaf plots;

(b) compute the five-point summaries and construct parallel boxplots;

(c) comment on your findings.

These data are available in data set `Intrusions`.

Solution. We denote the two sets with sorted numbers as

$$S_1 = \{37,\ 38,\ 43,\ 43,\ 47,\ 49,\ 50,\ 50,\ 54,\ 56,\ 56,\ 58,\ 59,\ 60\}$$

and

$$S_2 = \{21,\ 32,\ 32,\ 33,\ 35,\ 36,\ 36,\ 37,\ 38,\ 39,\ 39,\ 43,\ 44,\ 45,\ 45,\ 46,\ 48,\ 49,\ 53,\ 53\}.$$

Notice that there sample sizes are $|S_1| = 14$ and $|S_2| = 20$ respectively. Let x_i and y_j for $i = 1, \ldots, 14$ and $y_j = 1, \ldots, 20$ be the numbers listed in these two sets, respectively.

	2	1
7 8	3	2 2 3 5 6 6 7 8 9 9
3 3 7 9	4	3 4 5 5 6 8 9
0 0 4 6 6 8 9	5	3 3
0	6	

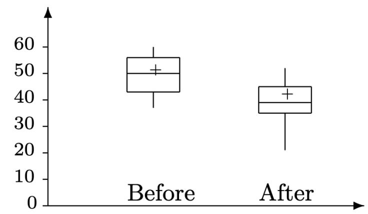

Figure 4: (Exercise 8.1) Left: Side-by-side stem-and-leaf plots of S_1 and S_2. Right: Parallel boxplots of S_1 and S_2.

(a) The stem-and-leaf plots of S_1 and S_2 are shown in Figure 4 left.

(b) The five-point summary of a data set is

$$(\min, \hat{Q}_1, \hat{M}, \hat{Q}_3, \max).$$

For S_1, there are $\min = x_1 = 37$ and $\max = x_{14} = 60$. Since $|S_1| = 14$, the sample median is $\hat{M} = \frac{x_7+x_8}{2} = \frac{50+50}{2} = 50$. Moreover, there are $25\% \times 14 = 3.5$ and $75\% \times 14 = 10.5$, and thus $\hat{Q}_1 = x_4 = 43$ and $\hat{Q}_3 = x_{11} = 56$. Therefore, the five-point summary of S_1 is

$$(37,\ 43,\ 50,\ 56,\ 60).$$

In addition, $\widehat{\text{IQR}} = \hat{Q}_3 - \hat{Q}_1 = 56 - 43 = 13$ and hence

$$\left[\hat{Q}_1 - 1.5(\widehat{\text{IQR}}),\ \hat{Q}_3 + 1.5(\widehat{\text{IQR}})\right] = [23.5, 75.5].$$

For S_2, there are $\min = y_1 = 21$ and $\max = y_{20} = 53$. Since $|S_2| = 20$, the sample median is $\hat{M} = \frac{y_{10}+y_{11}}{2} = \frac{39+39}{2} = 39$. Moreover, there are $25\% \times 20 = 5$ and $75\% \times 20 = 15$, and thus $\hat{Q}_1 = y_5 = 36$ and $\hat{Q}_3 = y_{15} = 46$. Therefore, the five-point summary of S_2 is

$$(21,\ 36,\ 39,\ 46,\ 53).$$

In addition, $\widehat{\text{IQR}} = \hat{Q}_3 - \hat{Q}_1 = 46 - 36 = 10$ and hence

$$\left[\hat{Q}_1 - 1.5(\widehat{\text{IQR}}),\ \hat{Q}_3 + 1.5(\widehat{\text{IQR}})\right] = [21, 61].$$

The parallel boxplots of S_1 and S_2 are shown in Figure 4 right.

(c) The sample median of S_1 is 50, and the sample median of S_2 is 39. The sample standard deviation of S_1 appears to be smaller than that of S_2 though.

After change of firewall settings, the number of blocked intrusion attempts appears to be smaller, but the variability appears to be larger.

8.4 Use Table A4 to compute the probability for any Normal random variable to take a value within 1.5 interquartile ranges from population quartiles.

Solution. Let X be the random variable that follows a normal distribution $\mathcal{N}(\mu, \sigma^2)$. Let Q_1 and Q_3 be the first and third quartiles of $\mathcal{N}(\mu, \sigma^2)$, respectively. Denote

$$z_i = \frac{Q_i - \mu}{\sigma}$$

for $i = 1, 3$. Notice that $z_1 \approx -0.67$ and $z_3 \approx 0.67$. Denote $X_{\widehat{\text{IQR}}} = Q_3 - Q_1$ and $Z_{\widehat{\text{IQR}}} = z_3 - z_1 \approx 1.34$.

Also notice that

$$\frac{X_{\widehat{\text{IQR}}}}{\sigma} = \frac{Q_3 - Q_1}{\sigma} = \frac{Q_3 - \mu}{\sigma} - \frac{Q_1 - \mu}{\sigma} = z_3 - z_1 = Z_{\widehat{\text{IQR}}}.$$

Therefore, we have

$$\begin{aligned}
P(Q_1 - 1.5X_{\widehat{\text{IQR}}} \le X \le Q_3 + 1.5X_{\widehat{\text{IQR}}}) \\
&= P\left(\frac{Q_1 - 1.5X_{\widehat{\text{IQR}}} - \mu}{\sigma} \le \frac{X - \mu}{\sigma} \le \frac{Q_3 + 1.5X_{\widehat{\text{IQR}}} - \mu}{\sigma}\right) \\
&= P\left(z_1 - 1.5\frac{X_{\widehat{\text{IQR}}}}{\sigma} \le Z \le z_3 + 1.5\frac{X_{\widehat{\text{IQR}}}}{\sigma}\right) \\
&= P(z_1 - 1.5Z_{\widehat{\text{IQR}}} \le Z \le z_3 + 1.5z_{\widehat{\text{IQR}}}) \\
&\approx P(-2.68 \le Z \le 2.68) \\
&= \Phi(2.68) - \Phi(-2.68) \\
&= 0.9963 - 0.0037 \\
&= 0.9926.
\end{aligned}$$

8.8 Consider three data sets (also, in data set Symmetry).

(1) 19, 24, 12, 19, 18, 24, 8, 5, 9, 20, 13, 11, 1, 12, 11, 10, 22, 21, 7, 16, 15, 15, 26, 16, 1, 13, 21, 21, 20, 19

(2) 17, 24, 21, 22, 26, 22, 19, 21, 23, 11, 19, 14, 23, 25, 26, 15, 17, 26, 21, 18, 19, 21, 24, 18, 16, 20, 21, 20, 23, 33

(3) 56, 52, 13, 34, 33, 18, 44, 41, 48, 75, 24, 19, 35, 27, 46, 62, 71, 24, 66, 94, 40, 18, 15, 39, 53, 23, 41, 78, 15, 35

(a) For each data set, draw a histogram and determine whether the distribution is right-skewed, left-skewed, or symmetric.

(b) Compute sample means and sample medians. Do they support your findings about skewness and symmetry? How?

Solution.

(a) The histograms of the three data sets are shown in Figure 5. The first data set appears to be left-skewed, the second data set appears to be symmetric, and the third data set appears to be right-skewed.

(b) The sample means and medians of the three data sets are shown in Table 1. Data set (1) has a mean of 14.97 and a median of 15.5 with difference of 0.5, which indicates that it is left-skewed. Data set (2) has a mean of 21.10 and a median of 21.0, which are very close with difference of 0.1, indicating that it is nearly symmetric. Data set (3) has a mean of 41.30 and a median of 39.5 with difference of 1.8, which indicates that it is right-skewed. These agree with the findings from the histograms.

8.9 The following data set represents the number of new computer accounts registered during ten consecutive days.

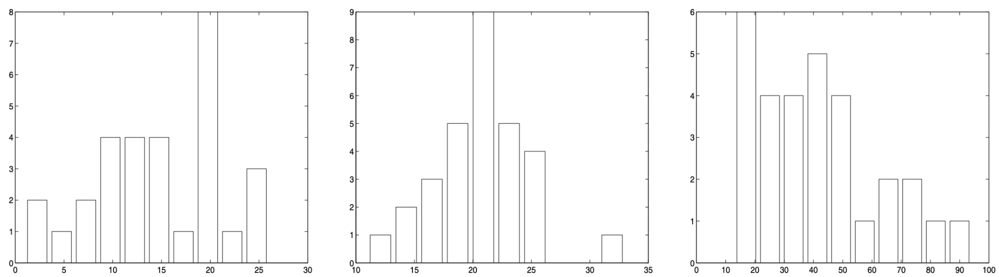

Figure 5: Histograms of the three data sets (1), (2), and (3) in Problem 8.8.

Table 1: Mean and median of the data sets in Problem 8.8.

Data Set	Mean $\bar{X}$	Median $\hat{M}$
(1)	14.97	15.5
(2)	21.10	21.0
(3)	41.30	39.5

$$43, 37, 50, 51, 58, 105, 52, 45, 45, 10.$$

(a) Compute the mean, median, quartiles, and standard deviation.

(b) Check for outliers using the 1.5(IQR) rule.

(c) Delete the detected outliers and compute the mean, median, quartiles, and standard deviation again.

(d) Make a conclusion about the effect of outliers on basic descriptive statistics.

These data are available in data set `Accounts`.

Solution.

(a) The sorted data set is

$$10, 37, 43, 45, 45, 50, 51, 52, 58, 105.$$

The mean is $\bar{X} = \frac{10+37+43+45+45+50+51+52+58+105}{10} = \frac{496}{10} = 49.6$. The median is $\hat{M} = \frac{45+50}{2} = 47.5$. The first quartile is $\hat{Q}_1 = 43$ and the third quartile is $\hat{Q}_3 = 52$. The interquartile range is $\widehat{\mathrm{IQR}} = \hat{Q}_3 - \hat{Q}_1 = 52 - 43 = 9$. The sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{X})^2 = 551.1556$$

and the standard deviation is $s = \sqrt{551.1556} \approx 23.46$.

(b) The lower bound is $\hat{Q}_1 - 1.5(\widehat{\mathrm{IQR}}) = 43 - 1.5(9) = 43 - 13.5 = 29.5$. The upper bound is $\hat{Q}_3 + 1.5(\widehat{\mathrm{IQR}}) = 52 + 1.5(9) = 52 + 13.5 = 65.5$. Thus, any data point below 29.5 or above 65.5 is considered an outlier. In this case, the outliers are 10 and 105.

(c) After removing the outliers, the new data set is

$$37, 43, 45, 45, 50, 51, 52, 58.$$

The new mean is $\bar{X} = \frac{37+43+45+45+50+51+52+58}{8} = \frac{381}{8} = 47.625$. The new median is $\hat{M} = \frac{45+50}{2} = 47.5$. The new first quartile is $\hat{Q}_1 = 43$ and the new third quartile is $\hat{Q}_3 = 52$. The new interquartile range is $\widehat{\text{IQR}} = \hat{Q}_3 - \hat{Q}_1 = 52 - 43 = 9$. The new sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{X})^2 = 49.0714$$

and the new standard deviation is $s = \sqrt{49.0714} \approx 7.00$.

(d) The five-point summary of the original data set is $(10, 43, 47.5, 52, 105)$ and the five-point summary of the new data set is $(37, 43, 47.5, 52, 58)$. The mean and standard deviation of the original data set are $\bar{X} = 49.6$ and $s \approx 23.46$, while the mean and standard deviation of the new data set are $\bar{X} = 47.625$ and $s \approx 7.00$. The outliers have a significant effect on the mean and standard deviation, and less effect on the median and quartiles.

Chapter 9

9.1 Estimate the unknown parameter θ from a sample

$$3,\ 3,\ 3,\ 3,\ 3,\ 7,\ 7,\ 7$$

drawn from a discrete distribution with the probability mass function

$$\begin{cases} P(3) = \theta \\ P(7) = 1 - \theta. \end{cases}$$

Compute two estimators of θ:

(a) The method of moments estimator;

(b) The maximum likelihood estimator;

(c) Estimate the standard error of each estimator of θ.

Solution.

(a) There is only one parameter and hence we just need to find it by solving $\mu_1 = m_1$, where the first moment (mean) of the PMF is

$$\mu_1 = \mathbb{E}(X) = 3 \cdot P(X = 3) + 7 \cdot P(X = 7) = 3\theta + 7(1 - \theta),$$

and the first sample moment is

$$m_1 = \bar{X}.$$

Solving $3\theta + 7(1-\theta) = \bar{X}$ for θ yields the estimator $\hat{\theta}_{\text{MoM}} = \frac{7-\bar{X}}{4}$.
Given the sample, we have $\bar{X} = \frac{3+3+3+3+3+7+7+7}{8} = \frac{36}{8} = 4.5$. Therefore, the value of $\hat{\theta}_{\text{MoM}}$ is

$$\hat{\theta}_{\text{MoM}} = \frac{7 - 4.5}{4} = \frac{2.5}{4} = 0.625.$$

(b) The likelihood function is

$$L(\theta) = \prod_{i=1}^{n} P(X = x_i) = \theta^{n_3}(1-\theta)^{n_7},$$

where n_3 is the number of observations equal to 3 and n_7 is the number of observations equal to 7.
The log-likelihood function is

$$\ell(\theta) = n_3 \log(\theta) + n_7 \log(1-\theta).$$

The maximum likelihood estimator $\hat{\theta}_{\text{MLE}}$ is obtained by solving the equation

$$\frac{d\ell(\theta)}{d\theta} = \frac{n_3}{\theta} - \frac{n_7}{1-\theta} = 0.$$

Solving this equation gives

$$\hat{\theta}_{\text{MLE}} = \frac{n_3}{n_3 + n_7}.$$

Since $n_3 = 5$ and $n_7 = 3$, we have

$$\hat{\theta}_{\text{MLE}} = \frac{5}{5+3} = \frac{5}{8} = 0.625.$$

(c) Notice that

$$\mathbb{E}(X) = 3\theta + 7(1-\theta) = 7 - 4\theta$$

and

$$\mathbb{E}(X^2) = 3^2\theta + 7^2(1-\theta) = 49 - 40\theta.$$

Therefore, the variance of X is

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = (49 - 40\theta) - (7 - 4\theta)^2 = 16\theta(1-\theta).$$

Hence the variance of $\hat{\theta}_{\text{MoM}}$ is

$$\text{Var}(\hat{\theta}_{\text{MoM}}) = \text{Var}(\frac{7 - \bar{X}}{4}) = \frac{1}{16}\text{Var}(\bar{X}) = \frac{1}{16} \cdot \frac{\text{Var}(X)}{n} = \frac{\theta(1-\theta)}{n}.$$

The standard error of $\hat{\theta}_{\text{MoM}}$ is

$$\text{SE}(\hat{\theta}_{\text{MoM}}) = \sqrt{\text{Var}(\hat{\theta}_{\text{MoM}})} = \sqrt{\frac{\theta(1-\theta)}{n}}.$$

We have $\hat{\theta}_{\text{MoM}} = 0.625$ as the estimate of θ and $n = 8$, so the estimated standard error is

$$\text{SE}(\hat{\theta}_{\text{MoM}}) \approx \sqrt{\frac{0.625(1-0.625)}{8}} = \sqrt{\frac{0.625 \cdot 0.375}{8}} \approx 0.1713.$$

Similarly, by solving the system of equations

$$\begin{cases} \bar{X} = \frac{3n_3 + 7n_7}{n_3 + n_7} \\ n = n_3 + n_7 \end{cases}$$

for n_3 and n_7, we can find that $\hat{\theta}_{\text{MLE}} = \frac{n_3}{n} = \frac{5}{8} = 0.625$. The variance of $\hat{\theta}_{\text{MLE}}$ is

$$\text{Var}(\hat{\theta}_{\text{MLE}}) = \frac{\theta(1-\theta)}{n}.$$

The standard error of $\hat{\theta}_{\text{MLE}}$ is

$$\text{SE}(\hat{\theta}_{\text{MLE}}) = \sqrt{\text{Var}(\hat{\theta}_{\text{MLE}})} = \sqrt{\frac{\theta(1-\theta)}{n}}.$$

We have $\hat{\theta}_{\text{MLE}} = 0.625$ as the estimate of θ and $n = 8$, so the estimated standard error is

$$\text{SE}(\hat{\theta}_{\text{MLE}}) \approx \sqrt{\frac{0.625(1-0.625)}{8}} = \sqrt{\frac{0.625 \cdot 0.375}{8}} \approx 0.1713.$$

9.2 The number of times a computer code is executed until it runs without errors has a Geometric distribution with unknown parameter p. For 5 independent computer projects, a student records the following numbers of runs:

3 7 5 3 2

Estimate p

(a) by the method of moments;

(b) by the method of maximum likelihood.

Solution.

(a) The first moment of the Geometric distribution is $\mu_1 = \frac{1}{p}$. The first sample moment is $m_1 = \bar{X}$. Solving $\mu_1 = m_1$ for p yields the estimator $\hat{p}_{\text{MoM}} = \frac{1}{\bar{X}}$. Given the sample, we have $\bar{X} = \frac{3+7+5+3+2}{5} = \frac{20}{5} = 4$. Therefore, the value of $\hat{p}_{\text{MoM}}$ is

$$\hat{p}_{\text{MoM}} = \frac{1}{4} = 0.25.$$

(b) The likelihood function is

$$L(p) = \prod_{i=1}^{n} P(X = x_i) = \prod_{i=1}^{n} p(1-p)^{x_i - 1} = p^n (1-p)^{\sum_{i=1}^{n}(x_i - 1)}.$$

The log-likelihood function is

$$\ell(p) = n\log(p) + \left(\sum_{i=1}^{n}(x_i - 1)\right)\log(1-p).$$

The maximum likelihood estimator $\hat{p}_{\text{MLE}}$ is obtained by solving the equation

$$\frac{d\ell(p)}{dp} = \frac{n}{p} - \frac{\sum_{i=1}^{n}(x_i - 1)}{1-p} = 0.$$

Solving this equation gives

$$\hat{p}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} x_i}.$$

Since $n = 5$ and $\sum_{i=1}^{n} x_i = 20$, we have

$$\hat{p}_{\text{MLE}} = \frac{5}{20} = 0.25.$$

9.3 Use method of moments and method of maximum likelihood to estimate

(a) parameters a and b if a sample from Uniform(a, b) distribution is observed;

(b) parameter λ if a sample from Exponential(λ) distribution is observed;

(c) parameter μ if a sample from Normal(μ, σ) distribution is observed, and we already know σ;

(d) parameter σ if a sample from Normal(μ, σ) distribution is observed, and we already know μ;

(e) parameters μ and σ if a sample from Normal(μ, σ) distribution is observed, and both μ and σ are unknown.

Solution.

(a) **Method of Moments.** For a Uniform(a, b) distribution, the probability density function is $f(x) = \frac{1}{b-a}$ for $a < x < b$. The first moment is $\mu_1 = \mathbb{E}(X) = \int_a^b f(x)dx = \int_a^b \frac{1}{b-a}dx = \frac{a+b}{2}$ and the second moment is $\mu_2 = \mathbb{E}(X^2) = \int_a^b x^2 f(x)dx = \int_a^b x^2 \frac{1}{b-a}dx = \frac{a^2+ab+b^2}{3}$. The first sample moment is $m_1 = \bar{X}$ and the second sample moment is $m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2$. Solving the system of equations $\mu_1 = m_1$ and $\mu_2 = m_2$ for a and b yields the method of moments estimators $\hat{a}_{\text{MoM}}$ and $\hat{b}_{\text{MoM}}$, which can be expressed as

$$\hat{a}_{\text{MoM}} = \bar{X} - \sqrt{3(m_2 - \bar{X}^2)}, \quad \hat{b}_{\text{MoM}} = \bar{X} + \sqrt{3(m_2 - \bar{X}^2)}.$$

Method of Maximum Likelihood. The likelihood function is

$$L(a, b) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \frac{1}{b-a} = \frac{1}{(b-a)^n}$$

for $a < X_i < b$ for all $i = 1, \ldots, n$. The log-likelihood function is

$$\ell(a, b) = -n\log(b-a).$$

where $a < X_i < b$ for all $i = 1, \ldots, n$. The maximum likelihood estimators $\hat{a}_{\text{MLE}}$ and $\hat{b}_{\text{MLE}}$ are obtained by maximizing the log-likelihood function subject to the constraints $a < X_i < b$ for all $i = 1, \ldots, n$. This leads to the maximum likelihood estimators are $\hat{a}_{\text{MLE}} = \min(X_1, \ldots, X_n)$ and $\hat{b}_{\text{MLE}} = \max(X_1, \ldots, X_n)$.

(b) **Method of Moments.** For an Exponential(λ) distribution, the first moment is $\mu_1 = \frac{1}{\lambda}$. The first sample moment is $m_1 = \bar{X}$. Solving $\mu_1 = m_1$ for λ yields the method of moments estimator $\hat{\lambda}_{\text{MoM}} = \frac{1}{\bar{X}}$.

Method of Maximum Likelihood. The likelihood function is

$$L(\lambda) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \lambda e^{-\lambda X_i} = \lambda^n e^{-\lambda \sum_{i=1}^n X_i}.$$

The log-likelihood function is

$$\ell(\lambda) = n\log(\lambda) - \lambda \sum_{i=1}^n X_i.$$

Taking the derivative of the log-likelihood function with respect to λ and setting it equal to zero gives

$$\frac{d\ell(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0.$$

Solving this equation for λ yields the maximum likelihood estimator

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\bar{X}}.$$

(c) **Method of Moments.** For a Normal(μ, σ) distribution with known σ, the first moment is $\mu_1 = \mu$. The first sample moment is $m_1 = \bar{X}$. Solving $\mu_1 = m_1$ for μ yields the method of moments estimator $\hat{\mu}_{\text{MoM}} = \bar{X}$.

Method of Maximum Likelihood. The likelihood function is

$$L(\mu) = \prod_{i=1}^{n} f(X_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X_i-\mu)^2}{2\sigma^2}}.$$

The log-likelihood function is

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - n\log(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Taking the derivative of the log-likelihood function with respect to μ and setting it equal to zero gives

$$\frac{d\ell(\mu)}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0.$$

Solving this equation for μ yields the maximum likelihood estimator

$$\hat{\mu}_{\text{MLE}} = \bar{X}.$$

(d) **Method of Moments.** For a Normal(μ, σ) distribution with known μ, the second moment is $\mu_2 = \sigma^2 + \mu^2$. The second sample moment is $m_2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$. Solving $\mu_2 = m_2$ for σ yields the method of moments estimator $\hat{\sigma}_{\text{MoM}} = \sqrt{m_2 - \mu^2}$.

Method of Maximum Likelihood. The likelihood function is

$$L(\sigma) = \prod_{i=1}^{n} f(X_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X_i-\mu)^2}{2\sigma^2}}.$$

The log-likelihood function is

$$\ell(\sigma) = -\frac{n}{2}\log(2\pi) - n\log(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Taking the derivative of the log-likelihood function with respect to σ and setting it equal to zero gives

$$\frac{d\ell(\sigma)}{d\sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

Solving this equation for σ yields the maximum likelihood estimator

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2}.$$

(e) **Method of Moments.** For a Normal(μ, σ) distribution with both μ and σ unknown, the first moment is $\mu_1 = \mu$ and the second moment is $\mu_2 = \sigma^2 + \mu^2$. The first sample moment is $m_1 = \bar{X}$ and the second sample moment is $m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2$. Solving the system of equations $\mu_1 = m_1$ and $\mu_2 = m_2$ for μ and σ yields the method of moments estimators $\hat{\mu}_{\text{MoM}} = \bar{X}$ and $\hat{\sigma}_{\text{MoM}} = \sqrt{m_2 - \bar{X}^2}$.

Method of Maximum Likelihood. The likelihood function is

$$L(\mu, \sigma) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X_i-\mu)^2}{2\sigma^2}}.$$

The log-likelihood function is

$$\ell(\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \mu)^2.$$

Taking the derivatives of the log-likelihood function with respect to μ and σ and setting them equal to zero gives the system of equations

$$\begin{cases} \frac{d\ell(\mu,\sigma)}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^n (X_i - \mu) = 0 \\ \frac{d\ell(\mu,\sigma)}{d\sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^n (X_i - \mu)^2 = 0 \end{cases}$$

Solving this system of equations for μ and σ yields the maximum likelihood estimators $\hat{\mu}_{\text{MLE}} = \bar{X}$ and $\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2}$.

9.5 A sample $(X_1, \ldots, X_{10})$ is drawn from a distribution with a probability density function

$$\frac{1}{2}\Big(\frac{1}{\theta}e^{-x/\theta} + \frac{1}{10}e^{-x/10}\Big), \quad 0 < x < \infty$$

The sum of all 10 observations equals 150.

(a) Estimate θ by the method of moments.

(b) Estimate the standard error of your estimator in (a).

Solution.

(a) The first moment of the given distribution is

$$\begin{aligned} \mu_1 = \mathbb{E}(X) &= \int_0^\infty x \cdot \frac{1}{2}\Big(\frac{1}{\theta}e^{-x/\theta} + \frac{1}{10}e^{-x/10}\Big) dx \\ &= \frac{1}{2}(\theta + 10) = \frac{\theta + 10}{2}. \end{aligned}$$

The first sample moment is $m_1 = \bar{X} = \frac{1}{10}\sum_{i=1}^{10} X_i = \frac{150}{10} = 15$. Solving $\mu_1 = m_1$ for θ yields the estimator $\hat{\theta}_{\text{MoM}} = 20$.

(b) The method of moments estimator is a linear function of the sample mean:

$$\hat{\theta}_{\text{MoM}} = 2\bar{X} - 10.$$

For the mixture distribution,

$$\mathbb{E}(X^2) = \frac{1}{2}(2\theta^2) + \frac{1}{2}(200) = \theta^2 + 100,$$

so

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 \\ &= \theta^2 + 100 - \left(\frac{\theta + 10}{2}\right)^2 \\ &= \frac{3\theta^2 - 20\theta + 300}{4}.\end{aligned}$$

Then

$$\mathrm{Var}(\hat{\theta}_{\mathrm{MoM}}) = 4\mathrm{Var}(\bar{X}) = \frac{4}{n}\mathrm{Var}(X) = \frac{3\theta^2 - 20\theta + 300}{10}.$$

Substituting $\theta = 20$ gives

$$\mathrm{Var}(\hat{\theta}_{\mathrm{MoM}}) = \frac{3(20)^2 - 20(20) + 300}{10} = \frac{1100}{10} = 110.$$

The standard error is

$$\mathrm{SE}(\hat{\theta}_{\mathrm{MoM}}) = \sqrt{110} \approx 10.49.$$

9.7 In order to ensure efficient usage of a server, it is necessary to estimate the mean number of concurrent users. According to records, the average number of concurrent users at 100 randomly selected times is 37.7, with a standard deviation $\sigma = 9.2$.

(a) Construct a 90% confidence interval for the expectation of the number of concurrent users.

(b) At the 1% significance level, do these data provide significant evidence that the mean number of concurrent users is greater than 35?

Solution.

(a) Since the sample size $n = 100$ is large, we can use the central limit theorem, which implies that the distribution of the sample mean is approximately normal, namely, $\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$. Therefore, the 90% confidence interval for the population mean is given by

$$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Here, $\bar{X} = 37.7$, $\sigma = 9.2$, $n = 100$, and $z_{\alpha/2} = z_{0.05} \approx 1.645$. Therefore, the confidence interval is

$$37.7 \pm 1.645 \cdot \frac{9.2}{\sqrt{100}} = 37.7 \pm 1.645 \cdot 0.92 \approx 37.7 \pm 1.5134.$$

Thus, the 90% confidence interval is approximately $(36.1866, 39.2134)$.

(b) To test the hypothesis $H_0 : \mu = 35$ against $H_A : \mu > 35$, we calculate the test statistic:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{37.7 - 35}{9.2/\sqrt{100}} = \frac{2.7}{0.92} \approx 2.935.$$

The critical value for a one-tailed test at the 1% significance level is $z_{0.01} \approx 2.33$. Since $z = 2.935 > 2.33$, we reject the null hypothesis. Therefore, there is significant evidence at the 1% level to conclude that the mean number of concurrent users is greater than 35.

9.8 Installation of a certain hardware takes random time with a standard deviation of 5 minutes.

(a) A computer technician installs this hardware on 64 different computers, with the average installation time of 42 minutes. Compute a 95% confidence interval for the population mean installation time.

(b) Suppose that the population mean installation time is 40 minutes. A technician installs the hardware on your PC. What is the probability that the installation time will be within the interval computed in (a)?

Solution.

(a) Since the sample size $n = 64$ is large, we can use the central limit theorem, which implies that the distribution of the sample mean is approximately normal, namely, $\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$. The 95% confidence interval for the population mean is given by

$$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Here, $\bar{X} = 42$, $\sigma = 5$, $n = 64$, and $z_{\alpha/2} = z_{0.025} \approx 1.96$. Therefore, the confidence interval is

$$42 \pm 1.96 \cdot \frac{5}{\sqrt{64}} = 42 \pm 1.96 \cdot \frac{5}{8} = 42 \pm 1.96 \cdot 0.625 \approx 42 \pm 1.225.$$

Thus, the 95% confidence interval is approximately $(40.775, 43.225)$.

(b) Since this refers to a single installation time X, and assuming $\mu = 40$ and $\sigma = 5$, we have $X \sim \mathcal{N}(40, 5^2)$ and

$$Z = \frac{X - 40}{5} \sim \mathcal{N}(0, 1).$$

Standardizing the endpoints gives

$$z_1 = \frac{40.775 - 40}{5} = 0.155, \qquad z_2 = \frac{43.225 - 40}{5} = 0.645.$$

Using the standard normal distribution,

$$P(Z < 0.155) = \Phi(0.155) \approx 0.5616, \qquad P(Z < 0.645) = \Phi(0.645) \approx 0.7401,$$

where $\Phi(z)$ is the cumulative distribution function of the standard normal distribution. Therefore,

$$P(40.775 < X < 43.225) = P(Z < 0.645) - P(Z < 0.155) \approx 0.7401 - 0.5616 = 0.1785.$$

9.9 Salaries of entry-level computer engineers have Normal distribution with unknown mean and variance. Three randomly selected computer engineers have salaries (in \$1000s):

30, 50, 70

(a) Construct a 90% confidence interval for the average salary of an entry-level computer engineer.

(b) Does this sample provide a significant evidence, at a 10% level of significance, that the average salary of all entry-level computer engineers is different from \$80,000? Explain.

(c) Looking at this sample, one may think that the starting salaries have a great deal of variability. Construct a 90% confidence interval for the standard deviation of entry-level salaries.

Solution.

(a) The sample mean is $\bar{X} = \frac{30+50+70}{3} = \frac{150}{3} = 50$. The sample standard deviation is

$$s = \sqrt{\frac{(30-50)^2 + (50-50)^2 + (70-50)^2}{3-1}} = \sqrt{\frac{400+0+400}{2}} = \sqrt{400} = 20.$$

Let μ denote the average salary of an entry-level computer engineer. Since the sample size $n = 3$ is small, we use the t-distribution to construct the confidence interval. The 90% confidence interval for μ is given by

$$\bar{X} \pm t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}.$$

Here, $\bar{X} = 50$, $s = 20$, $n = 3$, and $t_{\alpha/2,n-1} = t_{0.05,2} \approx 2.920$. Therefore, the confidence interval is

$$50 \pm 2.920 \cdot \frac{20}{\sqrt{3}} \approx 50 \pm 2.920 \cdot 11.547 \approx 50 \pm 33.71.$$

Thus, the 90% confidence interval is approximately $(16.29, 83.71)$.

(b) To test the hypothesis $H_0 : \mu = 80$ against $H_A : \mu \neq 80$, we calculate the test statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{50-80}{20/\sqrt{3}} = \frac{-30}{20/\sqrt{3}} = \frac{-30\sqrt{3}}{20} \approx -2.598.$$

The critical value for a two-tailed test at the 10% significance level with $n - 1 = 2$ degrees of freedom is $t_{0.05,2} \approx 2.920$. Since $|t| = 2.598 < 2.920$, we fail to reject the null hypothesis. Therefore, there is not enough evidence at the 10% level to conclude that the average salary of all entry-level computer engineers is different from $80,000.

(c) The 90% confidence interval for the population standard deviation is given by

$$\left(\sqrt{\frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}}, \sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}} \right).$$

Here, $n = 3$, $s^2 = 400$, and $\alpha = 0.10$. The critical values from the chi-square distribution with $n-1 = 2$ degrees of freedom are $\chi^2_{0.95,2} \approx 0.102$ and $\chi^2_{0.05,2} \approx 5.991$. Therefore, the confidence interval is

$$\left(\sqrt{\frac{(3-1)\cdot 400}{5.991}}, \sqrt{\frac{(3-1)\cdot 400}{0.102}} \right) \approx \left(\sqrt{\frac{800}{5.991}}, \sqrt{\frac{800}{0.102}} \right) \approx (11.55, 88.39).$$

9.10 We have to accept or reject a large shipment of items. For quality control purposes, we collect a sample of 200 items and find 24 defective items in it.

(a) Construct a 96% confidence interval for the proportion of defective items in the whole shipment.

(b) The manufacturer claims that at most one in 10 items in the shipment is defective. At the 4% level of significance, do we have sufficient evidence to disprove this claim? Do we have it at the 15% level?

Solution.

(a) The sample proportion of defective items is $\hat{p} = \frac{24}{200} = 0.12$. Since $n = 200$ is large, we can use the normal approximation $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$ and approximate the variance $p(1-p)/n$ by $\hat{p}(1-\hat{p})/n$. The 96% confidence interval for the population proportion is given by

$$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Here, $n = 200$, $\hat{p} = 0.12$, and $z_{\alpha/2} = z_{0.02} \approx 2.054$. Therefore, the confidence interval is

$$0.12 \pm 2.054 \cdot \sqrt{\frac{0.12(1-0.12)}{200}} \approx 0.12 \pm 2.054 \cdot \sqrt{\frac{0.12 \cdot 0.88}{200}} \approx 0.12 \pm 0.0472.$$

Thus, the 96% confidence interval is approximately $(0.0728, 0.1672)$.

(b) To test the hypothesis $H_0 : p \leq 0.1$ against $H_A : p > 0.1$, we calculate the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.12 - 0.1}{\sqrt{\frac{0.1(1-0.1)}{200}}} = \frac{0.02}{\sqrt{\frac{0.1 \cdot 0.9}{200}}} = \frac{0.02}{\sqrt{0.00045}} \approx \frac{0.02}{0.02121} \approx 0.943.$$

The critical value for a one-tailed test at the 4% significance level is $z_{0.04} \approx 1.751$. Since $z = 0.943 < 1.751$, we fail to reject the null hypothesis at the 4% level. Therefore, we do not have sufficient evidence to disprove the manufacturer's claim at the 4% level.

At the 15% significance level, the critical value is $z_{0.15} \approx 1.036$. Since $z = 0.943 < 1.036$, we also fail to reject the null hypothesis at the 15% level. Therefore, we do not have sufficient evidence to disprove the manufacturer's claim at the 15% level either.

9.12 An electronic parts factory produces resistors. Statistical analysis of the output suggests that resistances follow an approximately Normal distribution with a standard deviation of 0.2 ohms. A sample of 52 resistors has the average resistance of 0.62 ohms.

(a) Based on these data, construct a 95% confidence interval for the population mean resistance.

(b) If the actual population mean resistance is exactly 0.6 ohms, what is the probability that an average of 52 resistances is 0.62 ohms or higher?

Solution.

(a) Since the population standard deviation is known, the 95% confidence interval for the population mean is given by

$$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Here, $\bar{X} = 0.62$, $\sigma = 0.2$, $n = 52$, and $z_{\alpha/2} = z_{0.025} \approx 1.96$. Therefore, the confidence interval is

$$0.62 \pm 1.96 \cdot \frac{0.2}{\sqrt{52}} \approx 0.62 \pm 1.96 \cdot 0.0277 \approx 0.62 \pm 0.0543.$$

Thus, the 95% confidence interval is approximately $(0.5657, 0.6743)$.

(b) To find the probability that an average of 52 resistances is $\bar{X}$ =0.62 ohms or higher when the actual population mean is μ =0.6 ohms, we standardize the value:

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.62 - 0.6}{0.2/\sqrt{52}} = \frac{0.02}{0.0277} \approx 0.722.$$

Using the standard normal distribution, we find the probability:

$$P(Z \geq 0.722) = 1 - P(Z < 0.722) \approx 1 - 0.7642 = 0.2358.$$

9.16 A sample of 250 items from lot A contains 10 defective items, and a sample of 300 items from lot B is found to contain 18 defective items.

(a) Construct a 98% confidence interval for the difference of proportions of defective items.

(b) At a significance level $\alpha = 0.02$, is there a significant difference between the quality of the two lots?

Solution.

(a) The sample proportions of defective items are $\hat{p}_A = \frac{10}{250} = 0.04$ and $\hat{p}_B = \frac{18}{300} = 0.06$. The difference in sample proportions is $\hat{p}_A - \hat{p}_B = 0.04 - 0.06 = -0.02$. The standard error of the difference in proportions is given by

$$\begin{aligned} SE &= \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}} = \sqrt{\frac{0.04(1-0.04)}{250} + \frac{0.06(1-0.06)}{300}} \\ &\approx \sqrt{\frac{0.0384}{250} + \frac{0.0564}{300}} \approx \sqrt{0.0001536 + 0.000188} \approx \sqrt{0.0003416} \approx 0.01848. \end{aligned}$$

The 98% confidence interval for the difference in proportions is given by

$$(\hat{p}_A - \hat{p}_B) \pm z_{\alpha/2} \cdot SE.$$

Here, $z_{\alpha/2} = z_{0.01} \approx 2.326$. Therefore, the confidence interval is

$$-0.02 \pm 2.326 \cdot 0.01848 \approx -0.02 \pm 0.04296.$$

Thus, the 98% confidence interval is approximately $(-0.06296, 0.02296)$.

(b) To test the hypothesis $H_0 : p_A = p_B$ against $H_A : p_A \neq p_B$, we calculate the test statistic:

$$z = \frac{\hat{p}_A - \hat{p}_B}{SE} = \frac{-0.02}{0.01848} \approx -1.082.$$

The critical value for a two-tailed test at the 2% significance level is $z_{0.01} \approx 2.326$. Since $|z| = 1.082 < 2.326$, we fail to reject the null hypothesis. Therefore, there is not enough evidence at the 2% level to conclude that there is a significant difference between the quality of the two lots.

9.18 Consider the data about the number of blocked intrusions in Exercise 8.1, p. 240: The numbers of blocked intrusion attempts on each day during the first two weeks of the month were

$$56, 47, 49, 37, 38, 60, 50, 43, 43, 59, 50, 56, 54, 58$$

After the change of firewall settings, the numbers of blocked intrusions during the next 20 days were

$$53, 21, 32, 49, 45, 38, 44, 33, 32, 43, 53, 46, 36, 48, 39, 35, 37, 36, 39, 45.$$

(a) Construct a 95% confidence interval for the difference between the average number of intrusion attempts per day before and after the change of firewall settings (assume equal variances).

(b) Can we claim a significant reduction in the rate of intrusion attempts? The number of intrusion attempts each day has approximately Normal distribution. Compute P-values and state your conclusions under the assumption of equal variances and without it. Does this assumption make a difference?

Solution.

(a) Let $\bar{X}_1$ and $\bar{X}_2$ be the sample means of the number of blocked intrusions before and after the change of firewall settings, respectively. Let s_1^2 and s_2^2 be the sample variances. The 95% confidence interval for the difference in means is given by

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ is the pooled variance.
The sample mean and variance for the first group (before the change) are

$$\bar{X}_1 = 50.0, \qquad s_1^2 = 58, \qquad n_1 = 14.$$

The sample mean and variance for the second group (after the change) are

$$\bar{X}_2 = 40.2, \qquad s_2^2 = 63.3263, \qquad n_2 = 20.$$

The pooled variance is

$$s_p^2 = \frac{(14-1) \cdot 58 + (20-1) \cdot 63.3263}{14+20-2} \approx 61.1625.$$

The standard error of the difference in means is

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \approx \sqrt{61.1625} \cdot \sqrt{\frac{1}{14} + \frac{1}{20}} \approx 2.7252.$$

The critical value for a 95% confidence interval with $n_1 + n_2 - 2 = 32$ degrees of freedom is $t_{0.025,32} \approx 2.037$. Therefore, the confidence interval is

$$(50.0 - 40.2) \pm 2.037 \cdot 2.7252 \approx 9.8 \pm 5.55.$$

So the 95% confidence interval for the difference in means is approximately $(4.25, 15.35)$.

(b) To test the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ against $H_A : \mu_1 - \mu_2 > 0$, we calculate the test statistic:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE} \approx \frac{50.0 - 40.2}{2.7252} \approx 3.59.$$

The critical value for a one-tailed test at the 5% significance level with $n_1 + n_2 - 2 = 32$ degrees of freedom is $t_{0.05,32} \approx 1.694$. Since $t = 3.59 > 1.694$, we reject the null hypothesis. Therefore, there is significant evidence at the 5% level to conclude that the average number of intrusion attempts per day has decreased after the change of firewall settings.

Assuming that the number of intrusion attempts each day has approximately Normal distribution. The P-value for the test statistic $t = 3.59$ with $n_1 + n_2 - 2 = 32$ degrees of freedom is approximately 0.0005. Since the P-value is less than the significance level $\alpha = 0.05$, we reject the null hypothesis.